

Disorders of Sodium Balance

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CHAPTER OUTLINE

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KEY POINTS

- Sodium content of the extracellular compartment is a key determinant of circulatory integrity.
- Cumulative surfeits and deficits of body sodium, respectively, increase (hypervolemia) and decrease (hypovolemia) the extracellular fluid volume with attendant clinical consequences.
- Adjustments in the balance between filtered load and tubule reabsorption of sodium are responsible for many disorders of sodium balance.
- Hypervolemia and hypovolemia may be iso-, hypo-, or hyper-osmotic depending on the concomitant state of water balance.
- Sodium-retaining states include decompensated congestive heart failure, decompensated hepatic cirrhosis, and glomerular diseases.

Sodium (Na^+) and water balance and their distribution among the various body compartments are essential for the maintenance of fluid homeostasis, particularly intravascular volume. Disturbances of either or both components have serious medical consequences, are relatively frequent, and are among the most common conditions encountered in clinical practice. In fact, abnormalities of Na^+ and water balance are responsible for, or associated with, a wide spectrum of medical and surgical admissions or complications. The principal disorders of Na^+ balance are manifested clinically as hypovolemia or hypervolemia, whereas disruption in water balance can be diagnosed only in the laboratory as hyponatremia or hypernatremia. Although disorders of Na^+ and water balance are often interrelated, the latter are considered separately in Chapter 14. Here, the pathophysiologic features of Na^+ balance are discussed. Because Na^+ is restricted predominantly to the extracellular compartment, this chapter also addresses perturbations of extracellular fluid (ECF) volume homeostasis.

The preceding chapters in Section 1, which define the normal physiology and homeostatic processes of sodium and water balance, are a prerequisite for understanding pathophysiology. The kidneys play a key role in integrating

the interplay of local, systemic, hormonal, and neural factors, eventually translating into healthy handling of Na^+ (and other electrolytes) and water balance, or maladaptive Na^+ depletion or Na^+ retention.^{1,2} Basic concepts in physiology that will be referred to in pathophysiology are summarized in [Fig. 13.1](#) and [Tables 13.1 and 13.2](#) (see also in-depth discussion in previous versions of this chapter and recent reviews¹⁻³).

HYPOVOLEMIA

DEFINITION

Hypovolemia is the condition in which the volume of the ECF compartment is reduced in relation to its capacitance. The reduction may be absolute or relative. In states of absolute hypovolemia, Na^+ balance is truly negative, reflecting past or ongoing losses. Hypovolemia is described as “relative” when there is no Na^+ deficit but the capacitance of the ECF compartment is increased. In this situation of reduced effective arterial blood volume (EABV)—defined as that part of the ECF in the arterial blood system that

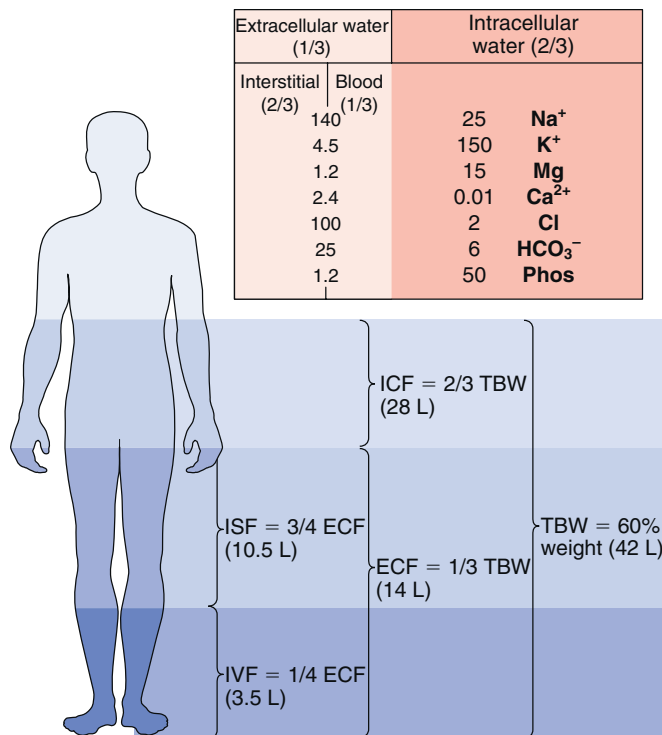


Fig. 13.1 Composition of body fluid compartments. This is a schematic representation of electrolyte composition (*upper panel*) and volumes (*lower panel*) of the body fluid compartments in humans. In the *upper panel*, electrolyte concentrations are in millimoles per liter; intracellular concentrations are typical values obtained from muscle. In the *lower panel*, shaded areas depict the approximate size of each compartment as a function of body weight. In a normally built individual, the total body water content is roughly 60% of body weight. Because adipose tissue has a low concentration of water, the relative water/total body weight ratio is lower in obese individuals. Relative volumes of each compartment are shown as fractions; approximate absolute volumes of the compartments (in liters) in a 70-kg adult are shown in parentheses. ECF, Extracellular fluid; ICF, intracellular fluid; ISF, interstitial fluid; IVF, intravascular fluid; TBW, total body water. (From Verbalis JG. Body water osmolality. In: Wilkinson B, Jamison R, eds. *Textbook of Nephrology*. London: Chapman & Hall; 1997:89–94. Reproduced with permission of Hodder Arnold.)

effectively perfuses the tissues—the ECF intravascular and extravascular (interstitial) compartments may vary in the same or opposite directions.

In physiologic terms, what is sensed is the threat to arterial pressure induced by the EABV⁴ that perfuses the arterial baroreceptors in the carotid sinus and glomerular afferent arterioles (see [Table 13.1](#)). Any change in perfusion pressure (or stretch) at these sites evokes appropriate compensatory responses in healthy individuals (see [Table 13.2](#)). EABV is usually correlated with actual ECF volume and is proportional to total body Na⁺. This means that regulation of Na⁺ balance and maintenance of EABV are closely related functions. Na⁺ loading generally leads to EABV expansion, whereas loss leads to depletion. However, in some situations, EABV and actual blood volume are not well correlated (see [Table 13.3](#)). For example, in heart failure (HF), a decrease in cardiac output leads to lowered perfusion pressure of the baroreceptors and reduced EABV is sensed. This leads to renal Na⁺ retention and ECF volume expansion. The net result is

Table 13.1 Mechanisms for Sensing Regional Changes in Effective Arterial Blood Volume

Sensors of Cardiac Filling

Atrial
Neural pathways
Humoral pathways
Ventricular
Pulmonary

Sensors of Cardiac Output

Carotid and aortic baroreceptors

Sensors of Organ Perfusion

Renal sensors
CNS sensors
GI tract sensors
Hepatic receptors
Guanylin peptides

CNS, Central nervous system; GI, gastrointestinal.

Table 13.2 Major Renal Effector Mechanisms for Regulating Effective Arterial Blood Volume

Glomerular Filtration Rate and Tubular Reabsorption

Tubuloglomerular feedback
Glomerulotubular balance
Peritubular capillary Starling forces
Luminal composition
Physical factors beyond proximal tubule
Medullary hemodynamics (pressure natriuresis)

Neural Mechanisms

Sympathetic nervous system
Renal nerves

Humoral Mechanisms

Renin-angiotensin-aldosterone system
Vasopressin
Prostaglandins
Natriuretic peptides
Endothelium-derived factors
Endothelins
Nitric oxide
Kinins
Adrenomedullin
Urotensin
Digitalis-like factors
Neuropeptide Y
Apelin
Glucagon-Like Peptide-1
Other novel factors^a

^aSee Leipziger J, Praetorius H. Renal autocrine and paracrine signaling: a story of self-protection. *Physiol Rev*. 2020;100:1229–1289.

Table 13.3 Causes of Absolute and Relative Hypovolemia

ABSOLUTE
Extrarenal
Gastrointestinal fluid loss
Bleeding
Skin fluid loss
Respiratory fluid loss
Extracorporeal ultrafiltration
Renal
Diuretics
Obstructive uropathy/postobstructive diuresis
Hormone deficiency
Hypoaldosteronism
Adrenal insufficiency
Na ⁺ wasting tubulopathies
Genetic
Acquired tubulointerstitial disease
RELATIVE
Extrarenal
Edematous states
Heart failure
Cirrhosis
Generalized vasodilation
Sepsis
Drugs
Pregnancy
Third-space loss
Renal
Severe nephrotic syndrome

increased plasma and total ECF volume, in association with increased EABV.⁵ The increase in plasma volume is partially appropriate in that intraventricular filling pressure rises leading to increasing myocardial stretching and improved ventricular contractility, thereby raising cardiac output and restoring systemic blood pressure (BP) and baroreceptor perfusion. However, this response is also maladaptive in that the elevated intraarterial and pulmonary pressures promote fluid movement out of the intravascular space into the tissues, which leads to peripheral and pulmonary edema, respectively.

In HF, EABV is dependent on cardiac output; in other disease settings, however, these two parameters may be dissociated. For example, in the presence of an arteriovenous fistula, cardiac output rises in proportion to the blood flow through the fistula. However, the flow through the fistula shunts blood away from the capillaries perfusing the tissues and therefore the EABV does not rise in conjunction with the rise in cardiac output. Similarly, a fall in systemic vascular resistance (SVR)—which, together with cardiac output, is a determinant of BP—leads to reductions in BP and EABV.

Another situation in which cardiac output and EABV change in opposite directions is advanced cirrhosis with ascites (see section, “Hypervolemia”).^{6,7}

Intracellular fluid (ICF) volume, reflected by measurements of plasma Na⁺ or osmolality, may or may not be concomitantly disturbed; thus hypovolemia may be classified as normonatremic, hyponatremic, or hypernatremic.

ETIOLOGY

The causes of hypovolemia are summarized in Table 13.3. Absolute and relative hypovolemia, in turn, can have extrarenal or renal causes. Absolute hypovolemia results from massive blood loss or fluid loss from the skin, gastrointestinal (GI), respiratory system, or kidneys. Relative hypovolemia results from states of vasodilation, generalized edema, or third-space loss. In both absolute and relative hypovolemia, the perceived reduction in EABV prompts compensatory hemodynamic changes and renal responses.

PATHOPHYSIOLOGY

ABSOLUTE HYPOVOLEMIA

Extrarenal

The commonest causes of absolute hypovolemia are persistent diarrhea, vomiting, and massive bleeding, either GI or as a result of trauma. The reduction in ECF volume is isotonic inasmuch as there is a proportionate loss of water and plasma. The consequent fall in systemic BP leads to compensatory tachycardia and vasoconstriction, and the ensuing altered transcapillary Starling hydraulic forces enable a shift of fluid from the interstitial to intravascular compartment. In addition, neural and hormonal responses to hypovolemia result in renal Na⁺ and water retention, with the aim of restoring intravascular volume and hemodynamic stability.

Similar compensatory mechanisms become activated after fluid losses from the skin, GI system, and respiratory system. Because of the large surface area of the skin, large amounts of fluid can be lost from this tissue because of burns or excessive perspiration. Severe burns cause the loss of large volumes of plasma and interstitial fluid and can lead rapidly to profound hypovolemia.⁸ Without medical intervention, hemoconcentration and hypoalbuminemia supervene. As occurs after massive bleeding, the fluid loss is isotonic, so plasma Na⁺ concentration and osmolality remain normal. In contrast, excessive sweating, induced by exertion in a hot environment, leads to hypotonic fluid loss because of the relatively low Na⁺ concentration (20 to 50 mmol/L) in sweat. The resulting hypovolemia may therefore be accompanied by hypernatremia and hyperosmolality and the type of fluid replacement must be tailored accordingly (see Chapter 14).⁹

In addition to oral intake, the GI tract is characterized by the entry of approximately 7 L of isotonic fluid, the overwhelming majority of which is reabsorbed in the large intestine. Hence in normal conditions, fecal fluid loss is minimal. However, in the presence of pathologic conditions, such as vomiting, diarrhea, and colostomy and ileostomy secretions, especially those caused by infection, considerable or even massive fluid loss may occur.¹⁰ The ionic composition, osmolality, and pH of secretions vary according to the part of the GI tract involved; therefore the resulting hypovolemia is associated with a large spectrum

of electrolyte and acid-base abnormalities (see [Chapters 15 and 16](#) for further discussion).

In contrast to the massive losses that can occur from the skin and GI system, fluid loss from the respiratory tract—as occurs in febrile states and in patients who receive mechanical ventilation with inadequate humidification—is usually modest and hypovolemia ensues only in the presence of accompanying causes. Finally, a special situation in which hypovolemia can occur is after excessive ultrafiltration in dialysis patients (see [Chapters 62 and 63](#)).

Renal

Even when glomerular filtration rate (GFR) is markedly impaired, the amount of filtered Na^+ far exceeds the dietary intake and all but ~1% of the filtered load is reabsorbed. However, if one or more of the tubular reabsorptive mechanisms are impaired, serious Na^+ deficit and absolute volume depletion can occur. The causes of absolute renal Na^+ losses include pharmacologic agents and renal structural, endocrine, and systemic disorders (see [Table 13.3](#)). All diuretics used to treat hypervolemic states may induce hypovolemia if administered in excess or inappropriately (see [Chapter 49](#)). Particularly, the powerful loop diuretics, furosemide, bumetanide, and torsemide, are often given in combination with diuretics acting on other tubular segments (e.g., thiazides, aldosterone antagonists, distal ENaC blockers, and carbonic anhydrase inhibitors). Patients receiving these combinations need to be carefully monitored and fluid balance should be scrupulously adjusted to prevent hypovolemia. Patients commonly at risk are those with HF or underlying hypertension who develop intercurrent infections.

In patients with hypertension, diuretic treatment appreciably increases the risk of volume depletion. Osmotic diuretics, endogenous or exogenous, may also reduce tubular Na^+ reabsorption. Endogenous agents include urea, the principal molecule involved in the polyuric recovery phase of acute kidney injury and postobstructive diuresis, and glucose in hyperglycemia. In patients with increased intracranial pressure, exogenous agents, such as mannitol or glycerol, may be used to induce translocation of fluid from the ICF to the ECF compartment and decrease brain swelling. The resulting polyuria may be associated with electrolyte and acid-base disturbances, the nature of which depends on the complex interplay between fluid intake and intercompartmental fluid shifts.

Na^+ reabsorption may also be disrupted in inherited and acquired tubular disorders. Inherited disorders of the proximal tubules (e.g., Fanconi syndrome) and distal tubules (e.g., Bartter and Gitelman syndromes) may lead to salt-wasting states in association with other electrolyte or acid-base disturbances. Acquired disorders of Na^+ reabsorption may be acute, as in nonoliguric acute kidney injury, the period immediately after renal transplantation, the polyuric recovery phase of acute kidney injury and postobstructive diuresis (please see, respectively, [Chapters 27, 28, 39, and 69](#)), or they may be chronic as a result of tubulointerstitial diseases with a propensity for salt wasting.

In addition to intrinsic tubular disorders, endocrine and other systemic disturbances may lead to impaired Na^+ reabsorption. The principal endocrine causes are mineralocorticoid deficiency and resistance states. A controversial cause is the systemic disturbance known as

cerebral salt wasting (CSW). In this condition, salt wasting is thought to occur in response to an as yet unidentified factor released in the setting of acute head injury or intracranial hemorrhage.^{11,12} CSW is usually diagnosed because of concomitant hyponatremia and signs of volume depletion in contrast to the normovolemia characteristic of the syndrome of inappropriate antidiuresis.¹³ Though one research group has proposed the designation of renal salt wasting and also accumulated single-center evidence that a signal peptide cleaved form of haptoglobin-related peptide drives natriuresis, the entity remains an enigmatic and not universally accepted clinical entity.¹³⁻¹⁵

An underappreciated, but not uncommon, clinical setting for renal Na^+ loss is after the administration of large volumes of intravenous saline to patients over several days after surgery or after trauma. In this situation, tubular reabsorption of Na^+ is downregulated.¹⁶⁻¹⁹ If intravenous fluids are stopped before full reabsorptive capacity is restored, volume depletion may ensue. The phenomenon can be minimized by graded reduction in the infusion rate, which allows Na^+ reabsorptive pathways to be restored gradually.

In the context of volume depletion, diabetes insipidus should be mentioned. However, because this results from a deficiency of or tubular resistance to arginine vasopressin (AVP), water loss is the main consequence and the impact on ECF volume is only minor. AVP-related disorders are considered in Chapter 14.

RELATIVE HYPOVOLEMIA

Extrarenal

As outlined previously, the principal causes of relative hypovolemia are edematous states, vasodilation, and third-space loss (see [Table 13.3](#)). Vasodilation may be physiologic, as in normal pregnancy, or induced by drugs (hypotensive agents, such as hydralazine or minoxidil, that cause arteriolar vasodilation), or it may occur in sepsis during the phase of peripheral vasodilation and consequent low SVR.²⁰

Edematous states in which the EABV and, hence, tissue perfusion are reduced include HF, decompensated cirrhosis with ascites, and nephrotic syndrome. In severe HF, low cardiac output and resulting low systemic BP lead to a fall in renal perfusion pressure (RPP). As in absolute hypovolemia, the kidneys respond by retaining Na^+ . Because the increased venous return cannot raise the cardiac output, a vicious cycle is created in which edema is further exacerbated and the persistently reduced cardiac output leads to further Na^+ retention. In decompensated cirrhosis, splanchnic venous pooling leads to decreased venous return, a consequent fall in cardiac output, and compensatory renal Na^+ retention. The pathophysiology of edematous states is discussed later (see section “Hypervolemia” later). Third-space loss occurs when fluid is sequestered into compartments not normally perfused with fluids, as in states of GI obstruction, after trauma, burns, or in pancreatitis, peritonitis, or malignant ascites. The result is that, even though total body Na^+ is markedly increased, the EABV is severely reduced.

Renal

Approximately 10% of patients with nephrotic syndrome—especially children with minimal change disease, but also any patient with a serum albumin level lower than 2 g/

dL—manifest clinical signs of hypovolemia.²¹ The low plasma oncotic pressure is conducive to movement of fluid from the ECF compartment to the interstitial space, thereby leading to reduced EABV.²²

CLINICAL MANIFESTATIONS

Clinical manifestations of hypovolemia depend on the magnitude and rate of volume loss, solute composition of the net fluid loss (i.e., the difference between input and output), and vascular and renal responses. Clinical features are related to the underlying pathophysiologic process, hemodynamic consequences, and electrolyte and acid-base disturbances accompanying the renal response to hypovolemia. A detailed history usually reveals the cause of volume depletion (bleeding, vomiting, diarrhea, polyuria, diaphoresis, medications).

Symptoms and physical signs of hypovolemia appear only when intravascular volume is decreased by 5% to 15% and are often related to tissue hypoperfusion. Symptoms include generalized weakness, muscle cramps, and postural light-headedness. Thirst is prominent if concomitant hypertonicity is present (hypertonic hypovolemia). Physical signs are related to the hemodynamic consequences of hypovolemia and include tachycardia; hypotension (postural, absolute or relative to the usual BP); and low central or jugular venous pressure. Elevated jugular venous pressure, however, does not rule out hypovolemia because of the possible confounding effects of underlying HF or lung disease. When volume depletion exceeds 10% to 20%, circulatory collapse is liable to occur, with severe supine hypotension, peripheral cyanosis, cold extremities, and impaired consciousness, extending even to coma. This is especially likely if fluid loss is rapid or occurs against a background of comorbid conditions. When the source of volume loss is extrarenal, oliguria also occurs. Traditional signs—reduced skin turgor, sunken eyes, and dry mucous membranes—are inconstant findings and their absence does not rule out hypovolemia. Reduction in the EABV, as manifested by relative hypotension, may also be observed in generalized edematous states, even though there is an overall excess of Na⁺ and water; however, this excess is maldistributed between the extracellular and interstitial spaces.

When hypovolemia occurs in the presence of arterial vasodilation, as observed in sepsis, some, but not all, of the clinical manifestations of hypovolemia are observed. Thus tachycardia and hypotension are usually present, but the extremities are warm, suggesting that perfusion is maintained. This finding is misleading because vital organs, particularly the brain and kidneys, are underperfused as a result of hypotension. The presence of lactic acidosis helps establish the correct diagnosis.²³

DIAGNOSIS

The diagnosis of hypovolemia is based essentially on clinical findings. Nevertheless, when these are equivocal, various laboratory parameters may be helpful for confirming the diagnosis or for elucidating other changes that may be associated with volume depletion.

LABORATORY FINDINGS

Hemoglobin and Plasma Albumin

Hemoglobin may decrease if significant bleeding has occurred or is ongoing, but the change, which is caused by hemodilution owing to fluid translocation from the interstitial to intravascular compartment, may take up to 24 hours. Therefore stable hemoglobin does not rule out significant bleeding. Moreover, the adaptive response of hemodilution may moderate the severity of hemodynamic compromise and resulting physical signs. In hypovolemic situations that do not arise from bleeding, hemoconcentration is often, but not universally, seen, inasmuch as underlying anemia of chronic disease may mask the differential loss of plasma.

Hemoconcentration may also be manifested as a rise in plasma albumin concentration if albumin-free fluid is lost from the skin, GI tract, or kidneys. On the other hand, when albumin is lost, either in parallel with other extracellular fluids (as in proteinuria, hepatic disease, protein-losing enteropathy, or catabolic states) or in protein-rich fluid (third-space sequestration, burns), significant hypoalbuminemia is observed.

Plasma Na⁺ Concentration

This may be low, normal, or high, depending on the solute composition of the fluid lost and the replacement solution administered by the patient or physician. For example, the hypovolemic stimulus for AVP release may lead to preferential water retention and hyponatremia, especially if hypotonic replacement fluid is used. In contrast, the fluid content of diarrhea may be hypotonic or hypertonic, resulting in hypernatremia or hyponatremia, respectively. The plasma Na⁺ concentration reflects the tonicity of plasma and provides no direct information about volume status, which is a clinical diagnosis.

Plasma K⁺ and Acid-Base Parameters

These can also change in hypovolemic conditions. After vomiting and after some forms of diarrhea, loss of K⁺ and Cl[−] may lead to metabolic alkalosis. More often, the principal anion lost in diarrhea is bicarbonate, which leads to hyperchloremic (nonanion gap) acidosis. When diuretics or Bartter and Gitelman syndromes (the inherited tubulopathies; see [Chapter 44](#)) are the cause of hypovolemia, hypokalemic alkalosis is again typically seen. On the other hand, urinary Na⁺ loss that occurs in adrenal insufficiency or due to aldosterone hyporesponsiveness is accompanied by a tendency for hyperkalemia and metabolic acidosis. Finally, when hypovolemia is sufficiently severe to impair tissue perfusion, high anion gap acidosis caused by lactic acid accumulation may be observed.

Blood Urea and Creatinine Levels

These frequently rise in hypovolemic states and reflect impaired renal perfusion. If tubular integrity is preserved, then the rise in urea levels is typically disproportionate to that of creatinine, so-called prerenal azotemia (see [Chapters 27 and 28](#)). This results mainly from AVP-enhanced urea reabsorption in the medullary collecting duct (MCD), but also from augmented proximal tubular reabsorption due to increased filtration fraction.²⁴ In

critically ill patients, an increased urea generation rate (from exogenous or endogenous protein catabolism) or low creatinine generation rate due to muscle wasting may lead to an erroneous diagnosis of prerenal azotemia.²⁵ In the presence of severe hypovolemia, acute kidney injury may ensue, leading to loss of the differential rise in urea level. Proportional rises in urea and creatinine are also observed when hypovolemia occurs against a background of underlying renal functional impairment, as in chronic kidney disease, stages 3 to 5.

Urine Biochemical Parameters

In hypovolemia due to extrarenal fluid losses, the intact kidney will respond to hypoperfusion by enhanced tubular reabsorption of Na^+ and water. The ensuing oliguria will be characterized by urine-specific gravity >1.020 , Na^+ concentration <10 mmol/L, and osmolality >400 mOsm/kg. When urine Na^+ concentration is 20 to 40 mmol/L, the finding of a fractional excretion of Na^+ of $<1\%$, in the presence of oliguria, may be helpful. However, in a patient with intrinsic tubular disease or injury or on previous diuretic therapy, especially with loop diuretics, these indices may merely reflect U_{Na} losses. In that case, fractional excretion of urea of $<30\%$ to 35% may help in the diagnosis of hypovolemia, although the specificity of this test is rather low.^{26,27}

Conversely, the presence of selective renal or glomerular ischemia (e.g., because of bilateral renal artery stenosis or acute glomerular injury) will be misinterpreted as poor renal perfusion and is associated with renal Na^+ retention (low U_{Na}).

TREATMENT

ABSOLUTE HYPOVOLEMIA

General Principles

The treatment goals of hypovolemia are to restore normal hemodynamic status and tissue perfusion. These goals are achieved by reversal of the clinical symptoms and signs, described previously. Treatment can be divided into three stages: 1. initial replacement of the immediate fluid deficit; 2. maintenance of the restored ECF volume in the presence of ongoing losses; and 3. treatment of the underlying cause, whenever possible. The main strategies to be addressed by the clinician are the route, volume, rate of administration, and composition of the replacement and maintenance fluids. These are liable to change according to the patient's response.

In general, when hypovolemia is associated with a significant hemodynamic disturbance, intravenous rehydration is required.^{28,29} (The use of oral electrolyte solutions in the management of infants and children is discussed in [Chapter 71](#)). The volume of fluid and rate of administration should be determined on the basis of the urgency of the threat to circulatory integrity, adequacy of the clinical response, and underlying cardiac function. Older patients are especially vulnerable to aggressive fluid challenge, and careful monitoring is required, particularly to prevent acute left ventricular failure and pulmonary edema from overzealous correction.

Sometimes the clinical signs do not point unequivocally to the diagnosis of hypovolemia, even though the history is strongly suggestive. Several tests and indices have been developed and validated. The use of dynamic measures is encouraged to help guide treatment. The gold standard is the invasive measure of cardiac output (or index) by thermodilution and its response to fluid challenge.³⁰ Responsiveness to volume could also be predicted using passive leg raising, end-expiratory occlusion, or minifluid challenge tests (comprehensively reviewed by Monnet and colleagues).³¹⁻³⁵ Invasive monitoring of central venous and pulmonary venous pressures has not been shown to improve outcomes in this situation; monitoring preload by stroke volume variation may improve outcomes, at least after major abdominal surgery.³⁶ Additionally, expanding the knowledge and practical application of noninvasive echocardiography and point-of-care ultrasound (POCUS, [Chapter 26](#)) could reliably assist clinicians in achieving the goals of volume management.^{30,31,37} Inferior vena cava (IVC) collapsibility index (CI) has been validated in both ventilated and nonventilated patients. An IVC maximum diameter <2.1 cm and IVC CI $>50\%$ are considered inconsistent with hypovolemia, while IVC CI $<20\%$ (with moderate to large IVC diameter) is inconsistent with intravascular volume depletion. Furthermore, combination of IVC ultrasound and lung ultrasound provides a more accurate assessment of intravascular and extravascular volume excess.³⁰

However, in case of doubt, a diagnostic fluid challenge should be performed. When a flat response to fluid challenge in any approved measure is achieved, further bolus volume therapy is futile and other therapies should be considered for shock management, such as vasopressors and correction of acidemia, among others.^{23,35}

In resource-limited countries or when dynamic measures are not available, good clinical judgment is essential for successful management because initial fluid deficits are difficult to calculate. Therefore patients with life-threatening circulatory collapse and hypovolemic shock require rapid intravenous replacement through the cannula with the widest bore possible. Replacement should continue until BP and tissue perfusion are restored. In the second stage, the rate of fluid replacement should be reduced to maintain BP and tissue perfusion. In older patients and those with underlying cardiac dysfunction, the risk of overrapid correction and precipitating pulmonary edema is heightened; therefore slower treatment is preferable to allow gradual filling of the ECF volume rather than causing pulmonary edema and the threat of mechanical ventilation associated with adverse outcomes.³⁸

Of note, evaluation and treatment of hypovolemia, electrolyte, and acid-base abnormalities associated with hyperglycemic crises (diabetic ketoacidosis and hyperglycemic hyperosmolar state) in adults with diabetes have been updated.³⁹ In patients without renal or cardiac compromise, the 2024 consensus report recommends starting the administration of isotonic saline or balanced solutions at an initial rate of 500 to 1000 mL/hour during the first 2 to 4 hours and correction of estimated deficits within the first 24 to 48 hours. Again, caution should be used in older patients with additional serious comorbidities and in pregnant individuals.

Composition of Replacement Fluids

The composition of replacement fluid may also affect outcomes.⁴⁰ The two main categories of replacement solutions are crystalloid and colloid solutions. Crystalloid solutions are based largely on NaCl of varying tonicity or dextrose. Isotonic (0.9%) saline, containing 154 mmol of Na⁺/L, is the mainstay of volume replacement therapy being confined to the ECF compartment in the absence of deviations in Na⁺ concentration. One L of isotonic saline increases plasma volume by approximately 300 mL; the rest is distributed to the interstitial compartment. In contrast, 1 L of 5% dextrose in water (D₅W), which is also isosmotic (277 mOsm/L), is eventually distributed throughout all the body fluid compartments so that only 10% to 15% (100–150 mL) remains in the ECF. Therefore D₅W should not be used for volume replacement.

Administration of 1 L of 0.45% saline (77 mmol of Na⁺/L) in D₅W is equivalent to giving 500 mL of isotonic saline and the same volume of solute-free water. The distribution of the solute-free compartment throughout all the fluid compartments would result in plasma dilution and reduction in the plasma Na⁺. Therefore this solution should be reserved for the management of hypernatremic hypovolemia. Even in that situation, it must be remembered that volume replacement is less efficient than with isotonic saline and, early in the treatment course, may cause plasma tonicity to fall too rapidly.

When hypovolemia is accompanied by severe metabolic acidosis (pH < 7.10; plasma HCO₃⁻ < 10 mmol/L), bicarbonate supplementation may be indicated. (For a discussion of bicarbonate balance, see [Chapter 15](#).) Because this anion is manufactured as 8.4% sodium bicarbonate (1000 mmol/L) for use in cardiac resuscitation, appropriate dilution is required for the treatment of acidosis associated with hypovolemia. Two convenient methods are suggested. Either 75 mL (75 mmol) of 8.4% NaHCO₃⁻ can be added to 1 L of .45% saline, or 150 mL of concentrated bicarbonate can be added to 1 L of D₅W. Although the latter is hypertonic in the short term, it is unlikely to be harmful.

In the presence of accompanying hypokalemia, especially if metabolic alkalosis is also present, volume replacement solutions should be supplemented with K⁺. Commercially available 1-L solutions of isotonic saline supplemented with 10 or 20 mmol of KCl make this option safe and convenient. (For details, see [Chapter 16](#).) On the other hand, newer, commercially available crystalloid solutions containing lactate (converted by the liver to bicarbonate) and low concentrations of KCl may offer advantages over isotonic saline. In a large prospective observational study performed in the intensive care unit setting, two periods were compared; in the control period, all patients received isotonic saline as fluid replacement, whereas during the intervention period, Hartmann solution (lactate-containing), Plasma-Lyte 148 (a balanced multielectrolyte solution, BMES), or chloride-poor 20% albumin solution was administered. The chloride-poor solutions were associated with a significantly lower risk of subsequent acute kidney injury, even after adjustment for covariates.⁴¹ Clearly, these provocative results indicated the need for well-controlled studies comparing BMES with chloride-rich solutions for fluid resuscitation.^{42,43} Indeed, two recent double-blind, randomized controlled trials (RCT) addressed this therapeutic dilemma in critically ill

patients.^{44,45} The PLUS study, an investigator-initiated, double-blind RCT, was conducted in 53 intensive care units (ICUs) in Australia and New Zealand.⁴⁴ The number of patients screened was 16,828, and about 30% were randomized (2515 and 2522 patients were assigned to receive BMES or saline, respectively). Compared with saline, the study showed that Plasma-Lyte 148 did not reduce 90-day all-cause mortality or risk of AKI and conferred no advantage in use of kidney replacement therapy (KRT) within 90 days, days alive and free of mechanical ventilation, or all-cause mortality at 28 days. These results were in line with the outcomes of the BaSICS trial, which was conducted in 75 ICUs in Brazil.⁴⁵ It was designed as a 2 × 2 RCT, assigning 10,520 critically ill patients into BMES (Plasma-Lyte 148) or 0.9% NaCl and a second randomization according to fluid rate delivery (333 mL/hour vs. 999 mL/hour). The 5290 patients were randomized to receive 0.9% NaCl; 2641 received the rapid infusion rate, whereas 5230 received BMES—2601 at rapid infusion rate. Results showed that there was no significant interaction between the two interventions, and neither demonstrated any significant difference between groups in primary outcome (90-day mortality) or secondary outcomes. However, in subgroup analysis, statistically significant interaction among presence of traumatic brain injury, fluid type, and 90-day mortality was evident (31.3% for the BMES group vs. 21.1% for the saline solution group [HR, 1.48; 95% CI, 1.03–2.12]).⁴⁵ Subsequently, two meta-analyses, based on data from six trials, suggested a high probability that the use of BMES in the ICU reduces in-hospital mortality, although the certainty of the evidence was moderate and the absolute risk reduction was small. Yet using BMES in the setting of traumatic brain injury was associated with increased in-hospital mortality.^{46,47}

Colloid solutions include plasma, albumin, and high-molecular-weight carbohydrate molecules, such as hydroxyethyl starch and dextrans, at concentrations that exert colloid oncotic pressure (COP) equal to or greater than that of plasma. Because the transcapillary barrier is impermeable to these large molecules, in theory, they expand the intravascular compartment more rapidly and efficiently than crystalloid solutions. Colloid solutions may be useful in the management of burns and severe trauma when plasma protein losses are substantial and rapid plasma expansion with relatively small volumes is efficacious. However, when capillary permeability is increased, as in states of multiorgan failure or the systemic inflammatory response syndrome, colloid administration is ineffective. Moreover, RCTs, in which crystalloid solutions were compared with colloid solutions, have shown no survival benefit and even harm with some colloid solutions, particularly hydroxyethyl starch.⁴³ Therefore the much cheaper and more readily available crystalloid solutions should remain the mainstay of therapy.

RELATIVE HYPOVOLEMIA

Treatment of relative hypovolemia is more difficult than that of absolute hypovolemia because there is no real fluid deficit. If the relative hypovolemia is caused by peripheral vasodilation, as in sepsis, it may be necessary to administer cautiously a crystalloid solution, such as isotonic saline, to maintain ECF volume until the SVR and venous capacitance

return to normal; the excess volume administered can then be excreted by the kidneys. For patients with sepsis-induced hypoperfusion or septic shock, the surviving sepsis campaign 2021 guidelines advocate immediate administration of 30 mL/Kg of intravenous crystalloid fluid, which should be given within the first 3 hours of resuscitation.²³ They also suggest using dynamic measures to guide fluid resuscitation, over physical examination or static parameters alone. Decrease in lactate level and capillary refill time was also suggested as an adjunct to other measures of perfusion. When vasodilation is more severe, vasoconstrictor agents may be needed. Current guidelines recommend an initial target mean arterial pressure (MAP) of 65 mm Hg over higher MAP targets.²³ In severe HF, advanced cirrhosis with portal hypertension and severe nephrotic syndrome, when EABV is low but there is an overall excess of Na⁺ and water, treatment may be extremely challenging. Crystalloid solution will, likely, lead to worsening interstitial edema without significantly affecting EABV. In these situations, prognosis is determined by whether the underlying condition can be reversed (see later).

HYPERVOLEMIA

DEFINITION

Hypervolemia occurs when the volume of the ECF compartment is expanded relative to its capacitance. Normally, increments in Na⁺ intake are matched by corresponding changes in Na⁺ excretion. However, in the approximately 20% of the population who are salt sensitive, the upward shift in ECF volume induced by high salt intake leads to a persistent rise in systemic arterial pressure, albeit without other signs of fluid retention (see [Chapter 46](#)).^{48,49} Here, the discussion is confined to hypervolemia, in which Na⁺ retention is ongoing and inappropriate for the prevailing ECF volume, with the appearance of clinical signs of volume overload.

ETIOLOGY

Hypervolemia may result from either primary renal Na⁺ retention or can be secondary to disease in other major organs ([Table 13.4](#)).

PRIMARY RENAL NA⁺ RETENTION

This can be subclassified as caused by intrinsic kidney disease or primary mineralocorticoid excess. Of the primary renal diseases causing Na⁺ retention, oliguric renal failure limits the ability to excrete Na⁺ and water, and affected patients are at risk for rapidly developing ECF volume overload (see [Chapters 27 and 28](#)). In contrast, in chronic kidney disease, renal tubular adaptation to salt intake is usually efficient until late-stage G4 and stage G5. However, in some primary glomerular diseases, especially in the presence of nephrotic-range proteinuria, significant Na⁺ retention may occur, even when GFR is close to normal (see section “Pathophysiology” later and [Chapter 30](#)). Primary mineralocorticoid excess leads to transient Na⁺ retention. However, because of “mineralocorticoid escape,” the dominant clinical feature is hypertension (see [Chapters 12 and 46](#)).

Table 13.4 Causes of Renal Sodium Retention

Primary

Oliguric acute kidney injury
Chronic kidney disease
Glomerular disease
Severe bilateral renal artery stenosis
Na⁺-retaining tubulopathies (genetic)
Mineralocorticoid excess

Secondary

Heart failure
Cirrhosis
Idiopathic edema

SECONDARY RENAL NA⁺ RETENTION

This occurs in low- and high-output cardiac failure with systolic and/or diastolic dysfunction. Nephrotic syndrome and hepatic cirrhosis with portal hypertension are also accompanied by renal Na⁺ retention. In this chapter, only HF and cirrhosis are considered. Nephrotic syndrome is discussed in [Chapter 30](#).

PATHOPHYSIOLOGY

Primary renal Na⁺ retention is caused by disruption of normal renal function. In contrast, secondary renal Na⁺ retention occurs because of reduced EABV in the presence of total ECF volume expansion or in response to factors secreted by the heart or liver that signal the kidneys to retain Na⁺ (see [Fig. 13.2](#)). In secondary Na⁺ retention, the renal effector mechanisms that normally operate to conserve Na⁺ and protect against a Na⁺ deficit are exaggerated and maintained, despite subtle or overt ECF volume expansion. The pathophysiology of hypervolemia involves local mechanisms of edema formation and stimulation of renal Na⁺ retention by reduced EABV either directly or indirectly, via abnormalities of the afferent volume sensing mechanisms.

LOCAL MECHANISMS OF EDEMA FORMATION

Peripheral interstitial fluid accumulation, which is common to all conditions causing hypervolemia, results from disruption of the normal balance of transcapillary Starling forces. Transcapillary fluid and solute transport consist of both convective and diffusive flow. Bulk water movement occurs via convective transport induced by hydraulic and osmotic pressure gradients. Capillary hydraulic pressure (P_c) is under the influence of several factors, including systemic arterial and venous BPs, local blood flow, and precapillary and postcapillary resistance. Systemic arterial BP, in turn, is determined by cardiac output, intravascular volume, and SVR; systemic venous pressure is determined by right atrial pressure, intravascular volume, and venous capacitance. Na⁺ balance is a key determinant of these latter hemodynamic parameters. Also, massive accumulation of fluid in the peripheral interstitial compartment (anasarca) can itself diminish venous compliance and, thereby, alter overall cardiovascular performance.⁵⁰

The balance of Starling forces prevailing at the arteriolar end of the capillary ($\Delta P > \Delta \pi$, in which $\Delta \pi$ is the change in transcapillary oncotic pressure) favors net filtration of fluid into the interstitium. Net outward movement of fluid along the length of the capillary is associated with an axial decrease in P_c and an increase in π_c . Nevertheless, the local ΔP continues to exceed the opposing $\Delta \pi$ throughout the length of the capillary bed in several tissues; thus filtration occurs along its entire length.⁵¹ In such capillary beds, a substantial volume of filtered fluid must therefore return to circulation via lymphatic vessels. Hence to minimize edema formation, the lymphatic vessels must be able to expand and proliferate and lymphatic flow increase in response to increased interstitial fluid formation.

Several other mechanisms for minimizing edema formation have been identified. First, precapillary vasoconstriction tends to lower P_c and diminish the filtering surface area in a given capillary bed. Indeed, in the absence of appropriate regulation of the microcirculatory myogenic reflex, as occurs with some Ca^{2+} channel blockers, excessive precapillary vasodilation may lead to lower-extremity interstitial edema.⁵² Second, increased net filtration itself is associated with dissipation of P_c , dilution of interstitial fluid protein concentration and a corresponding rise in intracapillary plasma protein concentration. The resulting change in the balance of Starling forces will tend to mitigate further interstitial fluid accumulation.⁵³ Finally, interstitial fluid hydraulic pressure (P_i) is normally subatmospheric; however, even small increases in interstitial fluid volume tend to augment P_i , again opposing further transudation of fluid into the interstitial space.⁵⁴ The appearance of generalized edema in association with expansion of the ECF volume therefore implies the presence of one or more disturbances in microcirculatory hemodynamics—increased venous pressure transmitted to the capillary, unfavorable adjustments in precapillary and postcapillary resistances, and/or inadequacy of lymphatic flow for draining the interstitial and replenishing the intravascular compartment.

For the clinical detection of generalized edema, the volume of accumulated interstitial fluid required (>2 to 3 L) necessitates expansion of ECF volume and, hence, body exchangeable Na^+ content. Since continued net accumulation of interstitial fluid without renal Na^+ retention might result in serious intravascular volume contraction and cessation of interstitial fluid formation, generalized edema must indicate substantial renal Na^+ retention.

SYSTEMIC FACTORS STIMULATING RENAL SODIUM RETENTION

Reduced Effective Arterial Blood Volume

Renal Na^+ (and water retention) in edematous disorders occurs due to reduced EABV, despite an increase in total blood and ECF volumes and normal intrinsic renal function.⁴ If the underlying stimulus for hypovolemia is removed, as dramatically seen after heart⁵⁵ or liver transplantation,⁵⁶ Na^+ excretion is restored to normal. Conversely, when kidneys from patients with end-stage liver disease are transplanted into patients with normal liver function, Na^+ retention no longer occurs.⁵⁶

Because 85% of blood circulates in the venous compartment, expansion of that compartment leads to

overall ECF volume excess that could occur concurrently with arterial underfilling. The latter could result from low cardiac output, peripheral arterial vasodilation, or a combination of the two. In turn, low cardiac output could result from true ECF volume depletion (see earlier discussion), cardiac failure, or decreased π_c , with or without increased capillary permeability. All these stimuli would cause activation of ventricular and arterial sensors. Similarly, conditions such as high-output cardiac failure, sepsis, cirrhosis, and normal pregnancy lead to peripheral arterial vasodilation and activation of arterial baroreceptors. Activation of these afferent mechanisms would then induce the neurohumoral mechanisms that result in renal Na^+ and water retention (see Fig. 13.2).⁵⁷

Although the mechanisms leading to Na^+ retention in HF and cirrhosis are similar, specific differences between the two conditions have been observed and are discussed separately in the following sections.

RENAL SODIUM RETENTION IN HEART FAILURE

Abnormalities of Sensing Mechanisms in Heart Failure

Both the cardiopulmonary and baroreceptor reflexes are blunted in HF, so they cannot exert an adequate tonic inhibitory effect on sympathetic outflow.⁵⁸ The resulting sympathetic nervous system (SNS) activation triggers, among others, renal Na^+ retention.^{5,59-61} A variety of models of HF have shown marked attenuation of atrial receptor firing and loss of nerve-ending arborization in HF.⁶² Similarly, altering central cardiac filling pressures in response to postural stimuli (e.g., head-up tilt and LBNP) in HF patients, in contrast to normal subjects, usually do not demonstrate significant alterations in limb blood flow, circulating catecholamines, AVP, or renin activity.^{63,64} This diminished reflex responsiveness is proportionate to the severity of ventricular dysfunction.

Arterial baroreceptor reflex impairment has been observed in HF. High baseline values of muscle sympathetic activity were found in patients with HF who failed to respond to activation and deactivation of arterial baroreceptors by infusion of phenylephrine and Na^+ nitroprusside, respectively.⁶² Carotid and aortic baroreceptor function were also depressed in experimental models of HF.⁶² These changes were associated with upward resetting of receptor threshold and a reduced range of pressures over which the receptors functioned.

Multiple abnormalities have been described in cardiopulmonary and arterial baroreceptor control of renal sympathetic nerve activity (RSNA) in HF. Thus rats with coronary ligation displayed an increased basal level of efferent RSNA that failed to decrease normally during volume expansion.^{65,66} Similarly, in sinoaortic denervated dogs with pacing-induced HF, or following left atrial baroreceptor stimulation, the cardiopulmonary baroreflex control of efferent RSNA was markedly attenuated.⁶⁷

The abnormal regulation of efferent RSNA was caused by impaired function of aortic and cardiopulmonary baroreflexes; the latter defect was functionally more important.⁶⁶ Mechanisms implicated in the pathogenesis of these abnormal baroreflexes include loss of compliance in the dilated hearts, gross changes in receptor structure,

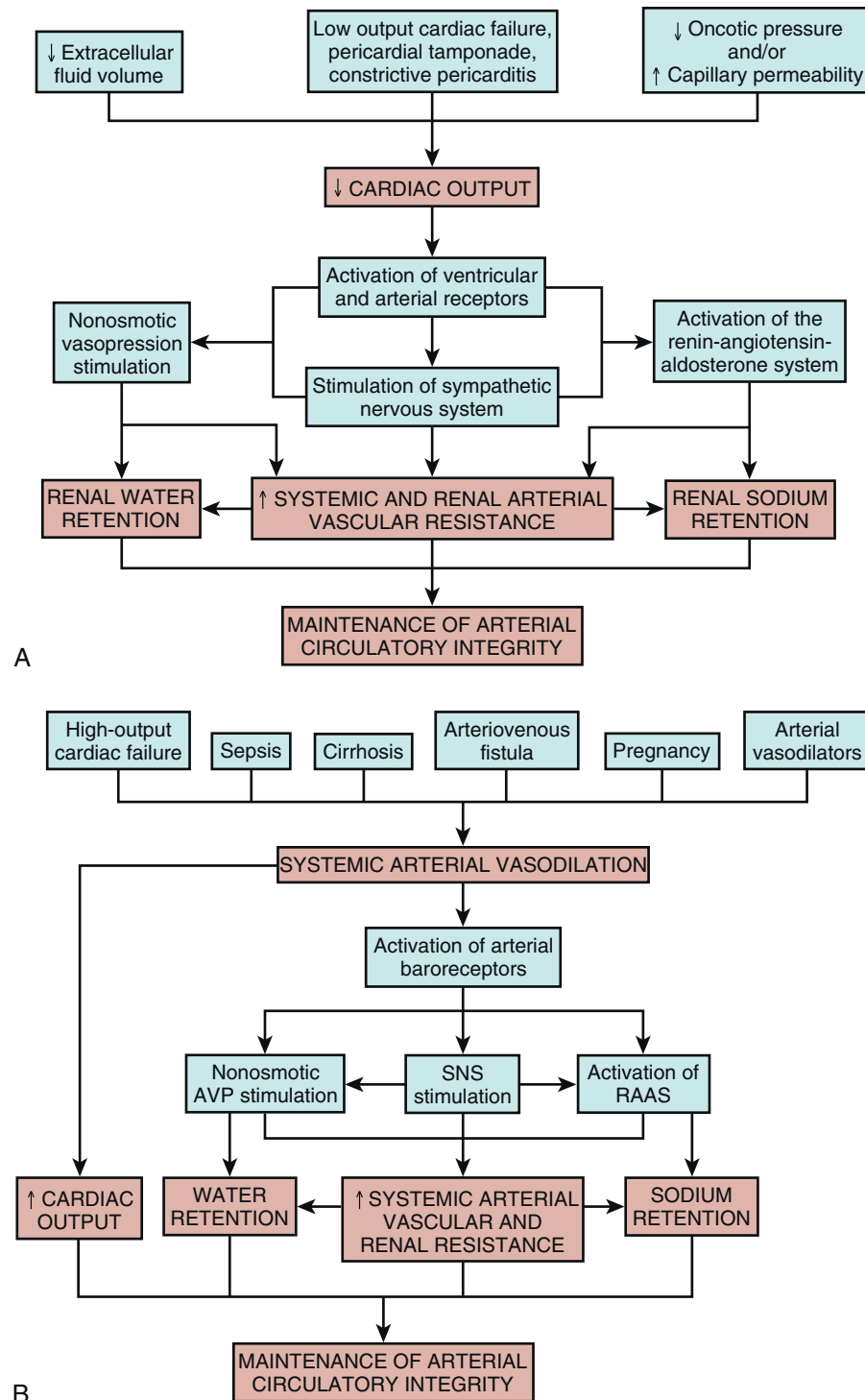


Fig. 13.2 Sensing mechanisms that initiate and maintain renal sodium and water retention in various clinical conditions in which arterial underfilling, with resultant neurohumoral activation and renal sodium and water retention, is caused by a decrease in cardiac output (A) and systemic arterial vasodilation (B). In addition to activating the neurohumoral axis, adrenergic stimulation causes renal vasoconstriction and enhances sodium and fluid transport by the proximal tubule epithelium. (From Schrier RW. Body fluid volume regulation in health and disease: A unifying hypothesis. *Ann Intern Med.* 1990;113:155–159.)

and augmented $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in the baroreceptor membranes.⁶² Increased activity of angiotensin II (Ang II) through the AT_1 receptor also contributes to depressed baroreflex sensitivity. Thus renin-angiotensin-aldosterone system (RAAS) inhibition in rats or rabbits with HF significantly improved arterial baroreflex control of RSNA

or heart rate, respectively.^{62,67} This effect of Ang II could also be blocked by central α_1 -adrenoreceptor stimulation.⁶⁸

Newer studies have indicated that Ang II in the paraventricular nucleus potentiates—and AT_1 receptor antisense mRNA normalizes—the enhanced cardiac sympathetic afferent reflex in rats with chronic HF.⁶²

AT₁ receptors in the nucleus tractus solitarius are thought to mediate the interaction between the baroreflex and cardiac sympathetic afferent reflex.⁶⁹ Consistent with this notion, Ang II generation is enhanced and its degradation is reduced in central sympathoregulatory neurons, as shown by upregulation of angiotensin-converting enzyme (ACE)1 and downregulation of ACE2.⁷⁰ AT₂ receptors in the rostral ventrolateral medulla inhibited sympathetic outflow, an effect mediated at least partly by an arachidonic acid metabolic pathway.⁷¹ These studies indicated that a downregulation in the AT₂ receptor was a contributory factor in the sympathetic neural excitation in HF.⁶²

Together, these data provide evidence of the role of high endogenous levels of Ang II, acting through the AT₁ receptor in concert with downregulation of the AT₂ receptor, in the impaired baroreflex sensitivity observed in HF, both in the afferent limb of the reflex arch and at more central sites. The central effect may be mediated through a central α_1 -adrenoreceptor. The blunted cardiopulmonary and arterial baroreceptor sensitivity in HF may also lead to an increase in AVP release and renin secretion.⁶²

The disturbances in sensing mechanisms that initiate and maintain renal Na⁺ retention in HF are summarized in Fig. 13.2A. As indicated, a decrease in cardiac output or a diversion of systemic blood flow diminishes the blood flow to the critical sites of the arterial circuit with pressure- and flow-sensing capabilities. The responses to diminished blood flow culminate in renal Na⁺ retention, mediated by the effector mechanisms. An increase in systemic venous pressure promotes the transudation of fluid from the intravascular to interstitial compartment by increasing the peripheral transcapillary ΔP . These processes augment the perceived loss of volume and flow in the arterial circuit. In addition, distortion of the pressure-volume relationships as a result of chronic dilation in the cardiac atria attenuates the normal natriuretic response to central venous congestion. This attenuation is manifested predominantly as a diminished neural suppressive response to atrial stretch, which results in increased sympathetic nerve activity and augmented release of renin and AVP.

Abnormalities of Effector Mechanisms in Heart Failure

The adaptive changes in the efferent limb of the volume control system in HF are generally similar to those seen in states of true Na⁺ depletion. These include adjustments in glomerular hemodynamics and tubular transport, brought about by alterations in the neural, humoral, and paracrine systems. However, in contrast to true Na⁺ depletion, HF is also associated with activation of vasodilatory/natriuretic agents, which tend to oppose the effects of the vasoconstrictor/antinatriuretic systems. The final effect on urinary Na⁺ excretion is determined by the dynamic balance among these antagonistic effector systems.

Alterations in Glomerular Hemodynamics

HF is characterized by not only increased renal vascular resistance and reduced GFR but also an even greater reduction in renal plasma flow (RPF), so the filtration fraction is increased.⁷² As shown in rat models of HF, these changes seem to result from diminished K_f and elevated afferent and efferent arteriolar resistances. The rise in

filtration fraction is probably caused by a disproportionate increase in efferent arteriolar resistance.⁷²

In Fig. 13.3, a comparison of the glomerular capillary hemodynamic profile in the normal versus the HF state is illustrated on the left graph of each panel. First, ΔP declines along the length of the glomerular capillary in normal and HF states, but much more so in HF because of the increased efferent arteriolar resistance. Second, $\Delta\pi$ increases over the length of the glomerular capillary in both states as fluid is filtered into the Bowman space, but again to a greater extent in HF because of the increased filtration fraction. As outlined later (see “Renin-Angiotensin-Aldosterone System” later), the preferential increase in efferent arteriolar resistance is mediated principally by Ang II and is critical for the preservation of GFR in the presence of reduced RPF. Because of the intense efferent arteriolar vasoconstriction, further compensation is not possible if RPP falls as a result of systemic hypotension, causing a sharp decline in GFR. This phenomenon is dramatically illustrated by HF patients whose Ang II drive is removed by RAAS inhibitors, particularly those with preexisting renal failure, massive diuretic treatment, and limited cardiac reserve.⁷² In these patients, BP may fall below the level necessary to maintain renal perfusion.

Enhanced Tubular Reabsorption of Sodium

A direct consequence of the glomerular hemodynamic alterations and augmented single-nephron filtration fraction is an increase in the fractional reabsorption of filtered Na⁺ in the proximal tubule. In Fig. 13.3, the peritubular capillary hemodynamic profile of the normal state is compared with that of HF on the right graph of each panel. In HF, in comparison with the normal state, the average value of $\Delta\pi$ along the peritubular capillary is increased and that of ΔP is decreased. These values favor fluid movement into the capillary and may also help reduce paracellular backleak of fluid into the tubule, promoting overall net reabsorption.

The peritubular control of proximal fluid reabsorption in normal and HF states is illustrated schematically in Fig. 13.3. A critical mediator of the enhanced tubular reabsorption of Na⁺ is Ang II, which, by increasing efferent arteriolar resistance, increases the filtration fraction and augments proximal epithelial transporter activities directly, thereby amplifying the overall increase in proximal Na⁺ reabsorption. This is clearly illustrated by the favorable effects of RAAS blockers in HF to modulate single-nephron filtration fraction and normalize proximal peritubular capillary Starling forces and Na⁺ reabsorption.⁷²

Enhanced reabsorption of Na⁺ in HF has been shown in the loop of Henle, probably due to altered renal hemodynamics, as in the proximal tubule.⁷³ In the distal tubule and collecting duct, elevated Ang II and aldosterone levels, respectively, enhance activities of the NaCl cotransporter and ENaC.⁷⁴

Neurohumoral Mediators

The primary vasoconstrictor/antinatriuretic (and antidiuretic) systems mediating Na⁺ and water retention in HF include the RAAS, SNS, AVP, and endothelins (ETs). The antagonistic vasodilator/natriuretic substances include nitric oxide (NO), prostaglandins (PGs), adrenomedullin (AM), urotensin II (UT II), and neuropeptide Y (NPY).

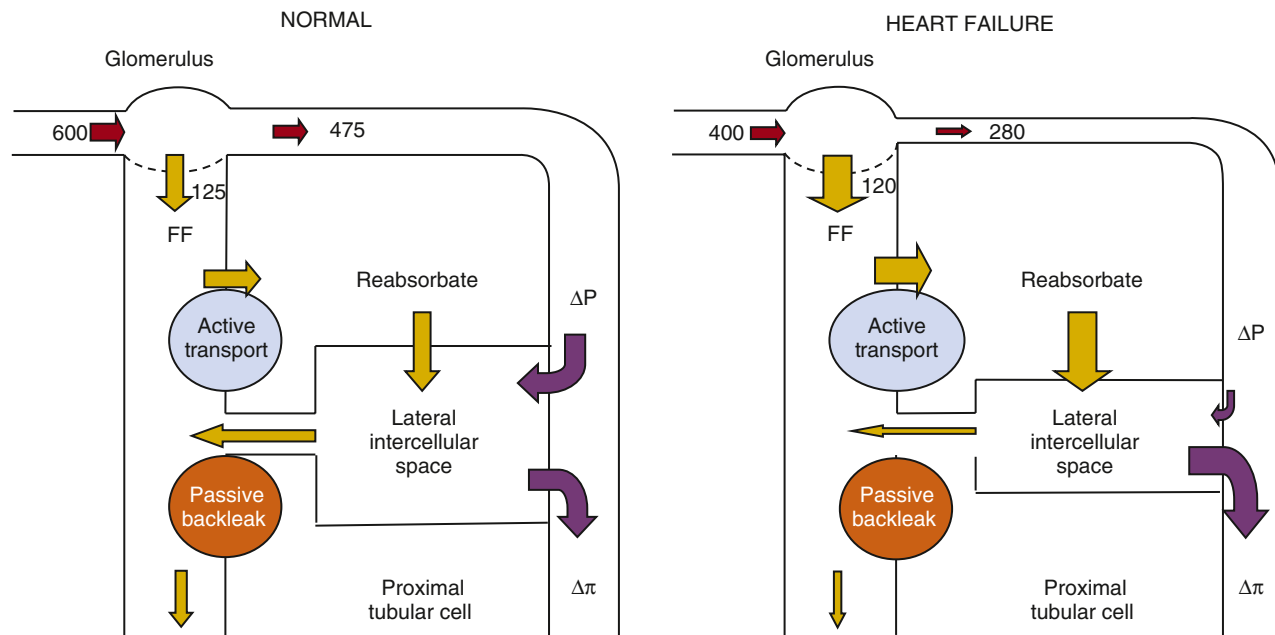


Fig. 13.3 Peritubular control of proximal tubule fluid reabsorption. Fluid reabsorption in the normal state (*left*) and in patients with heart failure (*right*) is shown. Increased postglomerular arteriolar resistance in heart failure is depicted as narrowing. The thickness and font size of the *block arrows* depict relative magnitude of effect. The increase in filtration fraction (FF) in heart failure causes $\Delta\pi$ to rise. The increase in renal vascular resistance in heart failure is believed to reduce ΔP . Both the increase in $\Delta\pi$ and fall in ΔP enhance peritubular capillary uptake of proximal reabsorbate and thus increase absolute Na^+ reabsorption by the proximal tubule. Numbers and *red block arrows* depict renal plasma flow [mL/min] in preglomerular and postglomerular capillaries; ΔP and $\Delta\pi$ are the transcapillary hydraulic and oncotic pressure differences across the peritubular capillary, respectively; *yellow block arrows* indicate transtubular transport; *purple block arrows* represent the effect of peritubular capillary Starling forces on uptake of proximal reabsorbate. (Modified from Humes HD, Gottlieb M, Brenner BM. *The Kidney in Congestive Heart Failure: Contemporary Issues in Nephrology*. Vol 1. New York; Churchill Livingstone; 1978:51–72.)

The development of positive Na^+ balance and edema in HF occurs at the point when the vasoconstrictor/antinatriuretic forces predominate (Fig. 13.4). The dominant activity of Na^+ -retaining systems in HF is clinically important, as it is associated with globally impaired renal function, a strong predictor of mortality⁷²; moreover, reversal of neurohumoral impairment is associated with improved outcomes.⁷²

VASOCONSTRICTOR/ANTINATRIURETIC (ANTIDIURETIC) SYSTEMS

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

The activity of the RAAS is enhanced in most patients with HF in correlation with the severity of cardiac dysfunction⁷⁵ and provides a prognostic index for HF patients. Initially, RAAS activation is beneficial by inducing direct systemic vasoconstriction and activating other neurohormonal systems such as AVP, which contribute to maintaining adequate intravascular volume.⁷⁵ However, numerous studies in patients and in experimental models of HF have established that continued activation of the RAAS leads to maladaptive myocardial remodeling⁷⁵ and progression of cardiovascular and renal dysfunction.³⁸

The kidneys in particular are highly sensitive to the action of Ang II, and a decrease in RPF and single-nephron glomerular filtration rate (GFR), as well as elevations in efferent arteriolar resistance and filtration fraction, are observed in both clinical and experimental HF. These changes are completely reversed by ACE inhibitors, as well as a low-salt diet.⁷⁶

Activation of Ang II in response to the decreased pumping capacity of the failing myocardium also promotes systemic vasoconstriction and mesangial cell contraction.⁷⁷ In addition, Ang II reduces renal cortical circulation in rats with HF and increases tubular Na^+ reabsorption directly and by augmenting aldosterone release.⁷⁷

Local RAAS in the heart and kidney is also important in maintaining Na^+ retention in HF. The phenomenon explains the presence of positive Na^+ balance, as well as the maintained efficacy of RAAS inhibition in chronic HF, in the absence of elevated systemic levels of the component hormones.⁷⁸ In general, it appears that systemic RAAS activation is most pronounced in acute decompensated HF, whereas local renal RAAS activation may dominate in chronic stable HF.

In the heart, local RAAS activation has a number of effects. In addition to the mechanical stress exerted on the myocardium due to systemic Ang II-mediated increased afterload, pressure overload activates local Ang II production as a result of upregulation of angiotensinogen and tissue ACE.⁷⁹ Local Ang II acts through AT_1 in a paracrine/autocrine manner, leading to cell swelling and cardiac hypertrophy, remodeling, and fibrosis (mediated by TGF- β), and reduced coronary flow, hallmarks of severe HF.⁷⁸ These observations explain the improved cardiac function, prolonged survival, prevention of end-organ damage, and prevention or regression of cardiac hypertrophy in HF treated with RAAS inhibitors.^{79,80} In addition, these drugs may improve endothelial dysfunction, vascular remodeling, and potentiation of the vasodilatory effects of kinins.⁸¹

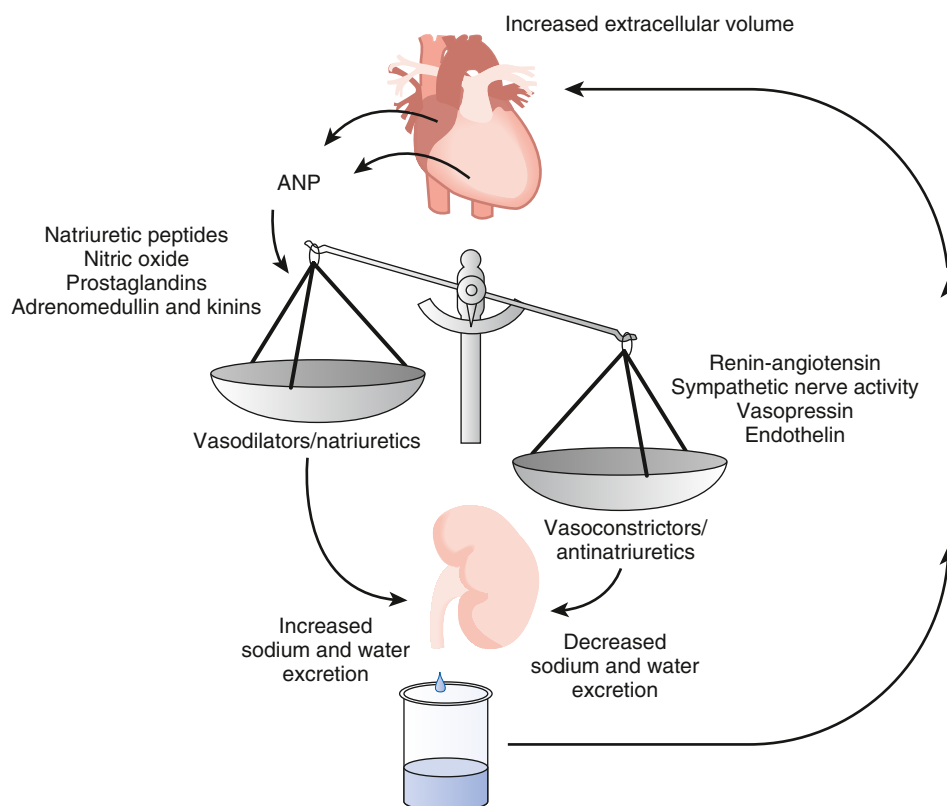


Fig. 13.4 Efferent limb of extracellular fluid volume control in heart failure. Volume homeostasis in heart failure is determined by the balance between natriuretic and antinatriuretic forces. In decompensated heart failure, enhanced activities of the Na^+ -retaining systems overwhelm the effects of the vasodilatory/natriuretic systems, which leads to a net reduction in Na^+ excretion and an increase in ECF volume. ANP, Atrial natriuretic peptide. (Modified from Winaver J, Hoffman A, Abassi Z, et al. Does the heart's hormone, ANP, help in congestive heart failure? *News Physiol Sci.* 1995;10:247–253.)

Like Ang II, aldosterone is produced locally and acts directly on the myocardium in HF inducing structural remodeling of the interstitial collagen matrix.⁸² These adverse effects of aldosterone were elegantly illustrated using eplerenone, a specific aldosterone antagonist, which prevented progressive left ventricular systolic and diastolic dysfunction by reducing interstitial fibrosis, cardiomyocyte hypertrophy, and left ventricular chamber sphericity in dogs with HF. Similarly, eplerenone attenuated ventricular remodeling and reactive (but not reparative) fibrosis after myocardial infarction in rats.^{83,84} These findings have been translated into the now routine clinical use of aldosterone antagonists in HF (see “Specific Treatments Based on the Pathophysiology of Heart Failure” later).⁸⁵

As noted, in addition to its renal and cardiovascular hemodynamic effects, the RAAS is involved directly in the exaggerated tubular Na^+ reabsorption in HF. Ang II, produced systemically and locally, directly stimulates proximal tubular Na^+ reabsorption.⁸⁶ In contrast, in the cortical and MCD, enhanced Na^+ reabsorption is mediated largely by aldosterone, as outlined previously. The pivotal role of aldosterone in HF is amply illustrated by elevated plasma and urine levels and the natriuretic effects of aldosterone antagonists in HF, despite further activation of other antinatriuretic systems.⁸⁷

The importance of RAAS action in the Na^+ retention of HF varies with the stage and severity of disease and, in more severe HF, positive Na^+ balance is associated with blunted

renal and hemodynamic responses to atrial natriuretic peptide (ANP); this response to ANP is restored by RAAS inhibition (for further details, see “Natriuretic Peptides” later).⁸⁸ Also, despite low plasma osmolality, patients with HF display increased thirst, probably because of the high Ang II concentrations, which stimulate thirst center cells in the hypothalamus.⁸⁹ This phenomenon may contribute to the positive water balance and hyponatremia often seen in advanced HF (see “Vasopressin” later).

SYMPATHETIC NERVOUS SYSTEM

As mentioned earlier, patients with HF experience progressive activation of the SNS with declining cardiac function^{75,58} and the adverse influence of sympathetic overactivity on the progression and outcome of patients with HF is abundantly clear.^{90,91} Thus plasma NE levels are frequently elevated and correlate with increased neural traffic. SNS activity is also significantly correlated with intracardiac pressures, cardiac hypertrophy, and left ventricular ejection fraction (LVEF).^{90,91} Activation of the SNS precedes the appearance of congestive symptoms and is also preferentially directed toward the heart and kidneys, as seen in patients with mild HF who have higher NE levels in the coronary sinus than in the renal veins.⁹¹ In early HF, increased SNS activity ameliorates the hemodynamic abnormalities including hypoperfusion, diminished plasma volume, and impaired cardiac function via vasoconstriction and avid Na^+ reabsorption.⁵⁸ However, chronic SNS activation induces several long-term adverse

myocardial effects including apoptosis and hypertrophy, with overall reduction in cardiac contractility. Some of these effects may be mediated by RAAS activation, which, in turn, can augment sympathetic activity and create a vicious cycle.⁷⁵

Basal sympathetic outflow to the kidneys is significantly increased in patients with HF,⁵⁸ and increased efferent RSNA contributes to the increased renal vasoconstriction, avid Na^+ and water retention, renin secretion, and attenuation of the renal actions of ANP.⁹² In rats with experimental HF caused by coronary artery ligation, renal denervation resulted in increased RPF and single-nephron GFR and decreased afferent and efferent arteriolar resistance.⁹⁰ In the same model, the decrease in RSNA in response to an acute saline load was less than that of control rats.⁶⁶ Conversely, bilateral renal denervation restored the natriuretic response to volume expansion. Similarly, in dogs with low cardiac output induced by vena caval constriction, administration of a ganglion blocker resulted in a marked increase in Na^+ excretion.⁹⁰ Also, in dogs with high-output HF induced by aortocaval fistula, total postprandial urinary Na^+ excretion was approximately twofold higher in dogs with renal denervation than in those with intact nerves.⁹⁰ In line with these observations, administration of the α -adrenoreceptor blocker dibenamine to patients with HF caused an increase in fractional Na^+ excretion, without a change in RPF or GFR. Treatment with ibopamine, an oral dopamine analog, resulted in vasodilation and positive inotropic and diuretic effects in these patients.⁹³ Moreover, for a given degree of cardiac dysfunction, the concentration of NE is significantly higher in patients with abnormal renal function than in those with preserved renal function.⁹⁴ These findings suggest that the association between renal function and prognosis in patients with HF is linked by both systemic and central nervous system neurohormonal activation.

RSNA may also affect renal hemodynamics and Na^+ excretion in HF by an antagonistic interaction with ANP. On the one hand, ANP has sympathoinhibitory effects^{95,96}; on the other hand, the SNS-induced salt and water retention in HF may reduce renal responsiveness to ANP. For example, the blunted diuretic/natriuretic response to ANP in rats with HF could be restored by prior renal denervation⁹⁷ or clonidine,⁹⁸ a centrally acting α_2 -adrenoreceptor agonist, which decreases RSNA in HF. These examples illustrate the complexity of interactions between the SNS and other humoral factors involved in the pathogenesis of Na^+ retention in HF.

In summary, the SNS plays an important role in the regulation of Na^+ excretion and glomerular hemodynamics in HF by either direct renal action or a complex interplay between the SNS itself and other neurohumoral mechanisms that act on the glomeruli and renal tubules. The introduction of renal denervation as a potential therapeutic treatment for HF should facilitate further elucidation of these neurohumoral interactions.

VASOPRESSIN

Numerous studies have demonstrated elevated plasma levels of AVP in HF, mostly in advanced HF with hyponatremia but also in asymptomatic patients with left ventricular dysfunction.⁹⁹ These high levels are related to nonosmotic factors such as attenuated left atrial compliance, hypotension,

and RAAS activation and are reversed by RAAS inhibition or α -blockade (prazosin).¹⁰⁰

The high circulating levels of AVP adversely affect the kidneys and cardiovascular system. In fact, raised levels of the C-terminal portion of the AVP prohormone (copeptin) at the time of diagnosis of acute decompensated HF are highly predictive of 1-year mortality.¹⁰¹ The prognostic power of an increased copeptin level in HF is similar to that of brain natriuretic peptide (BNP) levels (see “Brain Natriuretic Peptide” later). The most recognized renal effect of AVP in HF is the development of hyponatremia, especially in advanced stages of the disease, which most probably results from impaired solute-free water excretion, independent of plasma osmolality. In accordance with this notion, animal models of HF have demonstrated increased collecting duct expression of aquaporin 2 (AQP2).¹⁰² In addition, administration of specific V_2 receptor antagonists (VRAs) is associated with improvement in plasma Na^+ levels in animals and patients with hyponatremia.^{103,104} The improvement is associated with correction of the impaired urinary dilution in response to acute water load,¹⁰⁵ increased plasma osmolality, and downregulation of renal AQP2 expression but with no effect on renal blood flow (RBF), GFR, or Na^+ excretion.¹⁰⁶

The adverse effects of AVP on cardiac function¹⁰⁷ occur through its V_{1A} receptor to increase SVR (i.e., cardiac afterload), as well as by V_2 -receptor-mediated water retention, which leads to systemic and pulmonary congestion (increased preload). In addition, AVP, through its V_{1A} receptor, causes a direct rise in cardiomyocyte intracellular Ca^{2+} and activation of mitogen-activated kinases and protein kinase C. These signaling mechanisms appear to mediate the observed cardiac remodeling, dilation, and hypertrophy. The remodeling might be further exacerbated by the abnormalities in preload and afterload.

In summary, the data suggest that AVP is involved in the pathogenesis of water retention and hyponatremia that characterize HF and that AVP receptor antagonists result in remarkable diuresis in experimental and clinical models of HF. Treatment of HF with VRA is discussed further in “Specific Treatments Based on the Pathophysiology of Heart Failure”).

ENDOTHELIN

ET-1 is involved in both the development and progression of HF, as well as in the associated reduced renal function by inducing renal remodeling, interstitial fibrosis, glomerulosclerosis, hypoperfusion, hypofiltration, and positive salt and water balance.¹⁰⁸ The pathophysiologic role of ET-1 in HF is supported by two major lines of evidence: 1. The ET system is activated in HF; and 2. ET-1 receptor antagonists modify this pathophysiologic process.¹⁰⁸ The first line of evidence is based on elevated plasma ET-1 and big ET-1 concentrations in clinical HF and experimental models of HF; these levels correlate with hemodynamic severity and symptoms.¹⁰⁹ Also, the degree of pulmonary hypertension was the strongest predictor of plasma ET-1 level in patients with HF.¹¹⁰ Moreover, plasma levels of big ET and ET-1 are especially high in patients with moderate to severe HF and are independent markers of mortality and morbidity.¹⁰⁹ The increase in plasma ET-1 levels may be the result of enhanced synthesis in the lungs, heart,

and circulation by stimuli such as Ang II and thrombin or decreased pulmonary clearance.¹¹⁰ In parallel to ET-1 levels, ET-A receptors are upregulated, whereas ET-B receptors are downregulated in the failing human heart.¹¹¹

A cause-and-effect relationship between these hemodynamic abnormalities and ET-1 in HF was demonstrated using selective and highly specific ET receptor antagonists.¹¹⁰ In this regard, acute administration of the mixed ET-A/ET-B receptor antagonists, bosentan and tezosentan, significantly improved renal cortical perfusion, reversed the profoundly increased renal vascular resistance and increased RBF and Na⁺ excretion in rats with severe decompensated HF.¹¹² In addition, chronic blockade of ET-A by selective or dual ET-A/ET-B receptor antagonists attenuated the magnitude of Na⁺ retention and prevented the decline in GFR in experimental HF.¹¹³ These data are in line with earlier observations that rats with decompensated HF, as compared with normal rats, displayed severely blunted cortical vasoconstriction but significantly prolonged medullary vasodilation in response to ET-1 infusion. These effects could have resulted from activation of vasodilatory systems such as PG and NO, as exemplified by higher medullary immunoreactive endothelial nitric oxide synthase (eNOS) levels in rats with HF than controls.¹¹⁴ Taken together, the data indicate a role for ET in the pathogenesis of renal cortical vasoconstriction and Na⁺ retention in HF.

VASODILATORY/NATRIURETIC SYSTEMS

NATRIURETIC PEPTIDES

In decompensated HF, renal Na⁺ and water retention occur despite ECF volume expansion, even when the natriuretic peptide (NP) system is activated. Many clinical and experimental studies have implicated both ANP and BNP in the pathophysiology of the deranged cardiorenal axis in HF.

Atrial Natriuretic Peptide

Plasma levels of ANP and NH₂-terminal ANP are frequently elevated in HF and are correlated positively with the severity of cardiac failure, elevated atrial pressure, and left ventricular dysfunction.⁹² Hence circulating ANP level was proposed as a diagnostic marker of cardiac dysfunction and as a predictor of survival in HF.¹¹⁵ However, in this context, ANP has since been superseded by BNP. There is also evidence that mid-regional (MR) proANP may perform similarly to BNP as a biomarker of ADHF (see “Brain Natriuretic Peptide” later).¹¹⁶

The high plasma ANP levels are attributed to increased production rather than to decreased clearance. Although volume-induced atrial stretch is the main source for the elevated circulating ANP levels in HF, enhanced synthesis and release of the hormone by ventricular tissue in response to Ang II and ET also contribute to the elevated levels.⁹² Despite the high levels, patients and experimental animals with HF retain salt and water because renal responsiveness to NPs is attenuated.¹¹⁷ However, infusion of ANP to patients with HF does lead to hemodynamic improvement and inhibition of activated neurohumoral systems. These data are in line with findings that ANP is a weak counterregulator of the vasoconstriction mediated by the SNS, RAAS, and AVF.¹¹⁸ However, despite the blunted renal response to ANP in HF, elimination of ANP production by atrial

appendectomy in dogs with HF aggravated the activation of the vasoconstrictive factors and resulted in marked Na⁺ and water retention.¹¹⁹ These data suggest that ANP plays a critical role in suppressing Na⁺-retaining systems and as an important adaptive or compensatory mechanism aimed at reducing pulmonary vascular resistance and hypervolemia.

Brain Natriuretic Peptide

Plasma levels of BNP and N-terminal (NT)-proBNP are elevated in severe HF in proportion to the degree of myocardial systolic and diastolic dysfunction and the New York Heart Association (NYHA) classification.^{92,120} The extreme elevation of plasma BNP in severe HF stems mainly from increased synthesis by the hypertrophied ventricular tissue, but the atria also contribute.^{92,120}

Although echocardiography remains the gold standard for the evaluation of left ventricular dysfunction, plasma levels of BNP and NT-proBNP are reliable markers and, in fact, superior to ANP and NT-proANP for the diagnosis and prognosis of HF.¹²¹ NT-proBNP, in particular, has high sensitivity, specificity, and negative predictive value in patients with an ejection fraction <35%. Similar high predictive values are found in patients with concomitant left ventricular hypertrophy, either in the absence of or after myocardial infarction.¹²¹ The added presence of renal dysfunction appears to enhance these predictive values,^{122,123} and graded increases in mortality throughout each quartile of BNP levels have been shown in several clinical trials.¹²¹ In addition, elevated plasma BNP (or NT-proBNP) levels and LVEF lower than 40% are complementary independent predictors of death, HF, and new myocardial infarction at 3 years after a first infarction. Moreover, risk stratification with the combination of LVEF lower than 40% and high levels of NT-proBNP is substantially better than that provided by either alone.¹²⁴ However, even though BNP levels tend to be lower in patients with preserved LVEF than in HF patients with reduced LVEF, the prognosis in patients with preserved LVEF is as poor as in those with reduced LVEF for a given BNP level.¹²⁴

In asymptomatic patients with preserved LVEF, elevated BNP levels are correlated with diastolic abnormalities in Doppler studies. Conversely, a reduction in BNP levels with treatment is associated with a reduction in left ventricular filling pressures, lower readmission rates, and a better prognosis; thus monitoring of BNP levels may provide valuable information regarding treatment efficacy and expected patient outcomes.¹²⁵

Another diagnostic role for BNP is in the distinction of dyspnea caused by HF from that caused by noncardiac diseases. Using specific cut points, NT-proBNP levels are highly sensitive and specific for the diagnosis of acute HF. Levels lower than 300 pg/mL rule out acute HF, with a negative predictive value of 99%. An increased level of NT-proBNP is the strongest independent predictor of a final diagnosis of acute HF. NT-proBNP testing alone was superior to clinical judgment, the National Health and Nutrition Examination score, and Framingham clinical parameters alone for diagnosing acute HF; NT-proBNP plus clinical judgment was superior to NT-proBNP or clinical judgment alone.¹²¹

There is also evidence, albeit of rather low quality, that circulating BNP and NT-proBNP levels can be useful as a

guide to therapeutic efficacy of drugs typically prescribed in HF including RAAS inhibitors, diuretics, digitalis, and β -blockers.^{126,127} Also, BNP, but not NT-proBNP, levels at 24 and 48 hours after admission for acute decompensated HF (ADHF) predicted both 30-day and 1-year mortality. Predischage levels of both peptides were predictive of 30-day and 1-year mortality but not 1-year readmission due to HF.¹²⁸ In contrast, BNP levels were not helpful in reducing length of hospital stay and costs.¹²⁶

Together, these findings suggest that a simple and rapid determination of plasma levels of BNP or NT-proBNP in HF patients, together with clinical and echocardiographic measures, can be used to assess cardiac dysfunction, serve as a diagnostic and prognostic marker, and possibly assist in titrating relevant therapy.¹²¹ However, it should be emphasized that plasma NP levels are affected by age, salt intake, gender, obesity, hemodynamic status, and renal function leading to considerable overlap among diagnostic groups.^{121,129} Measurement of a panel may enhance the ability of biomarkers to distinguish between cardiac and noncardiac causes of dyspnea.¹³⁰

C-Type Natriuretic Peptide

Like those of ANP and BNP, plasma C-type NP (CNP) levels are increased in HF and are directly correlated with NYHA classification; levels of BNP, ET-1, and AM; pulmonary capillary wedge pressure; ejection fraction; and left ventricular end-diastolic diameter.¹³¹ CNP is synthesized mainly in the kidney,¹³² but it is also processed by the myocardium. Overexpression of CNP in the myocardium during HF may be involved in counteracting cardiac remodeling.¹³¹ On the other hand, renal CNP secretion is blunted in HF.¹³³ In contrast to the diminished physiologic responses to ANP and BNP in animals with HF, CNP elicited twice as much soluble guanylate cyclase (sGC) activity as ANP due to dramatic reductions in NP receptor-A (NPR-A) but not NPR-B activity.¹³¹ These findings imply a significant role for NPR-B-mediated NP activity in HF and may explain the modest effects of the NPR-A-selective nesiritide (BNP) treatment in HF.¹³⁴

Overall, current evidence points to a role of CNP in either the peripheral vascular compensatory response in HF or mitigation of the cardiac remodeling characteristic of HF. Elaboration of the exact role of CNP in HF appears crucial for the design of more effective NP analogs than those currently available for the management of HF.

OVERALL RELATIONSHIP BETWEEN NATRIURETIC AND ANTINATRIURETIC FACTORS IN HEART FAILURE

The maintenance of Na^+ balance in the initial compensated phase of HF is at least in part due to the elevated ANP and BNP levels.⁹² This notion is supported by the findings that, in experimental HF, inhibition of NP receptors by specific antibodies increased renal vascular resistance and decreased GFR, RBF, urine flow, Na^+ excretion, and RAAS activation.¹³⁵ In addition, NPs inhibited the Ang II-induced systemic vasoconstriction, proximal tubule Na^+ reabsorption, and secretion of aldosterone and ET.¹³⁶

In view of the remarkable activation and ability of NPs to counter the vasoconstrictor/antinatriuretic neurohormonal effects, why then do salt and water retention occur in overt HF? Several mechanisms could explain this paradox:

1. Appearance of abnormal circulating peptides and inadequate secretory reserves in comparison with the degree of HF. Using an extremely sensitive mass spectrometry-based method, altered processing of proBNP1-108 and/or BNP1-32 has been demonstrated, resulting in very low levels of BNP1-32, despite markedly elevated levels of immunoreactive (i.e., total) BNP.¹³⁷ Moreover, proBNP1-108 has a lower affinity for the guanylate cyclase A receptor, which would reduce effector function of BNP.¹³⁸
2. Decreased availability of NPs by downregulation of corin¹³⁹ or upregulation of NEP and clearance receptors.¹³⁵ With respect to corin, lower plasma levels have been observed in HF patients in parallel with elevated levels of proANP. On the other hand, circulating cGMP levels are elevated in HF, implying enhanced activity of NPs. These apparently contradictory findings could be reconciled by the demonstration of low intracardiac corin levels in an experimental model of HF caused by dilated cardiomyopathy. Moreover, transfection of the gene encoding for corin into these animals led to a reduction in cardiac fibrosis, improvement in contractility, and reduced mortality.¹³⁹ Regarding clearance receptors, there is no convincing evidence to date of upregulation in the renal tissue of HF animals or patients, although increased abundance of clearance receptors for NPs in platelets of patients with advanced HF has been reported.¹⁴⁰ In contrast, enhanced expression and activity of NEP in experimental HF are well documented⁹² and NEP inhibitors improve the vascular and renal response to NPs in HF (see “Specific Treatments Based on the Pathophysiology of Heart Failure” later).
3. Activation of vasoconstrictor/antinatriuretic factors and renal hyporesponsiveness to ANP. Renal resistance to ANP may be present, even in the early presymptomatic stage of the disease and progresses proportionately as HF worsens.¹³⁵ In advanced HF, when RPF is markedly impaired, the ability of NPs to antagonize the renal effects of the maximally activated RAAS is limited.⁹² The mechanisms underlying the attenuated renal effects of ANP in HF include Ang II-induced afferent and efferent vasoconstriction, mesangial cell contraction, activation of cGMP phosphodiesterases that attenuate the accumulation of the second messenger of NPs in target organs, and stimulation of Na^+ - H^+ -exchanger and Na^+ channels in the proximal tubule and collecting duct, respectively.¹⁴¹
4. Activation of the SNS also can overwhelm the renal effects of ANP. As described earlier, overactivity of the SNS leads to vasoconstriction of the peripheral circulation and of the afferent and efferent arterioles, which causes reduction of RPF and GFR. These actions, together with the direct stimulatory effects of SNS on Na^+ reabsorption in the proximal tubule and loop of Henle, contribute to the attenuated renal responsiveness to ANP in HF. Moreover, the SNS-induced renal hypoperfusion/hyofiltration stimulates renin secretion, thereby aggravating the positive Na^+ and water balance. In rat models of HF, the natriuretic responses to ANP were increased after sympathetic inhibition by low-dose clonidine⁹⁸ or bilateral renal denervation.¹⁴² The beneficial effects of renal denervation could be attributed to upregulation of NP receptors and cGMP production.¹⁴³

In summary, the development of renal hyporesponsiveness to NPs is paralleled closely by overreactivity of the RAAS and SNS and represents a critical point in the development of positive salt balance and edema formation in advanced HF.

NITRIC OXIDE

NO is implicated in the increased vascular resistance and impaired endothelium-dependent vascular responses characteristic of HF.^{144,145} The impaired activity is mediated by reduced shear stress associated with the decreased cardiac output, downregulation or uncoupling of eNOS, decreased availability of the NO precursor L-arginine caused by increased activity of arginase, and increased levels of the endogenous NOS inhibitor asymmetric dimethyl arginine (ADMA). In addition, inactivation of NO by superoxide ion and alteration of the redox state of sGC through oxidative stress lead to reduced levels of the NO-sensitive form of sGC and, thereby, of its second messenger cGMP.^{144,145} Oxidative stress may be further exacerbated by overactivity of counterregulatory neurohumoral systems, such as the RAAS and the release of proinflammatory messengers.^{144,145}

Altered activity of the NO-sGC-cGMP system also underlies the regional vasomotor dysregulation of the renal circulation in HF, and Ang II may be involved in mediating the impaired NO-dependent renal vasodilation.¹⁴⁶ The resulting imbalance between NO and excessive activation of the RAAS and ET systems could explain some of the beneficial effects of RAAS inhibition.¹⁴⁷ Support for this imbalance concept came from a model of experimental HF in rats overexpressing eNOS in the renal medulla and, to a lesser extent, in the cortex.¹¹⁴ This eNOS might play a role in the preservation of intact medullary perfusion and could attenuate severe cortical vasoconstriction. Accumulation of ADMA as exemplified by elevated plasma levels in normotensive HF could also account for the impaired renal hemodynamics in HF. In fact, in a multiple regression analysis, ADMA levels independently predicted reduced RBF.¹⁴⁸

Locally generated NO by the myocardium is also believed to modulate cardiac function and thereby lead to impaired renal function in HF.^{144,145} Alterations in the expression of cardiac NOS isoforms in HF are complex, and the functional consequences of these changes depend on a balance among various factors including disruption of the unique subcellular localization of each isoform and nitroso-redox imbalance.^{144,145}

In summary, endothelium-dependent vasodilation is attenuated in various vascular beds in HF. This attenuation may occur as a result of decreased NO levels and downregulation or inhibition of downstream NO signal transduction pathways. These effects may occur directly or via counterregulatory vasoconstrictor neurohumoral mechanisms.

PROSTAGLANDINS

PGs play an important role in maintaining renal function in the setting of impaired RBF in HF. Renal hypoperfusion, directly or by RAAS activation, stimulates the release of PGs that exert a vasodilatory effect, predominantly on the afferent arteriole, and promote Na⁺ excretion by inhibiting reabsorption in the thick ascending limb of Henle and the MCD.^{149,150} Evidence for the compensatory role of PG

in experimental and clinical HF comes from two sources. First, plasma levels of PGE₂, PGE₂ metabolites, and 6-keto-PGF₁ were higher in HF patients than in normal subjects.¹⁵¹ Moreover, in both experimental and human HF, there is a direct relationship between plasma renin activity (PRA)/Ang II and plasma and urinary PGE₂ and PGI₂ metabolite concentrations.¹⁵² This correlation probably reflects both Ang II-induced stimulation of PG synthesis and PG-mediated increased renin release. A similar counterregulatory role of PG regarding the other vasoconstrictors (e.g., catecholamines and AVP) may also be inferred.

The second approach, which established the protective role of renal and vascular PG in HF, was by using nonsteroidal antiinflammatory drugs (NSAIDs) to inhibit PG synthesis. In various experimental models of HF, this maneuver was associated with elevated urinary PGE₂ excretion, increase in body weight, and renal vascular resistance with resultant decrease in RBF, mainly due to afferent arteriolar constriction.^{151,153} Urine flow rate declined significantly and serum creatinine and urea rose.¹⁵³ Similarly, in patients with HF and hyponatremia, in whom extreme activation of the SNS and RAAS occurred, significant decreases in RBF and GFR accompanied by reduced urinary Na⁺ excretion, followed by NSAID treatment.^{151,154} These effects were prevented by intravenous PGE₂. Moreover, pretreatment with indomethacin attenuated the captopril-induced increase in RBF.¹⁵⁴ Thus ACEI-associated improvement in renal hemodynamics is mediated in part by increased PG synthesis.

Selective cyclooxygenase-2 (COX-2) inhibitors also lead to a significant worsening of chronic HF and renal function, especially in older patients taking diuretics.^{155,156,157} These deleterious effects are predictable given the relative abundance of COX-2 in renal tissue and, to a lesser extent, the myocardium of HF patients.^{149,158}

In summary, HF can be viewed as a PG-dependent state, in which elevated Ang II levels and enhanced RSNA stimulate renal synthesis of PGE₂ and PGI₂, to counteract the vasoconstrictor neurohumoral stimuli and maintain GFR and RBF. Both COX-2 and nonselective COX inhibitors should be avoided in HF because they leave the vasoconstrictor systems unopposed, leading to hypoperfusion, hypofiltration, and Na⁺ and water retention.¹⁴⁹

ADRENOMEDULLIN

AM seems to play a role in the pathophysiology of HF. HF patients have up to fivefold elevations in plasma levels of AM, in proportion to the severity of cardiac, hemodynamic, and neurohumoral derangements including pulmonary arterial and capillary wedge pressure, NE, ANP, BNP levels, and PRA.^{159,160} Plasma levels of AM decreased with effective anti-HF treatment, such as carvedilol.¹⁶¹ High levels of mid-regional proAM are also strong predictors of mortality in HF.^{162,163,164} The origin of the increased circulating AM appears to be the failing ventricular and, to a lesser extent, atrial myocardium.^{161,165}

Both cardiac and renal AM levels are significantly increased in some, although not all, experimental models of HF.^{166,167} The renal upregulation is consistent with the favorable acute and more prolonged (4 days) effects of AM on creatinine clearance, Na⁺, and water excretion, as well as on the hemodynamic abnormalities of experimental HF.¹⁶¹

In contrast, acute administration of AM to HF patients increased forearm blood flow but less so than in normal subjects. Stroke index and dilation of resistance arteries were increased, and plasma aldosterone was reduced, but Na^+ and water excretion were unaffected.¹⁶¹ Collectively, the data suggest that AM acts to balance the elevation in SVR and volume expansion in HF.¹⁵⁹

Because the favorable effects of AM alone are rather modest, combination therapy with other vasodilatory/natriuretic substances has been attempted. Combinations with BNP, ACEIs, NEP inhibitors, and epinephrine resulted in hemodynamic and renal benefits greater than those achieved by each agent alone.^{161,168} A small long-term clinical trial of combined ANP and AM in acute decompensated HF demonstrated a significant increase in cardiac output, reductions in MAP, pulmonary arterial pressure, and systemic and pulmonary vascular resistance without changing heart rate. In addition, levels of aldosterone, BNP, and free radical metabolites fell and Na^+ and water excretion rose.¹⁶⁹

With the advent of neutral endopeptidase (NEP) inhibitors, which enhance the activities of vasodilatory/natriuretic peptides, including AM, HF outcomes have been significantly improved. (See “Specific Treatments Based on the Pathophysiology of Heart Failure.”)

UROTENSIN

A role for UT II and its G protein–coupled receptor, GPR14, or UT II receptor, in the pathogenesis of HF has been suggested. First, some, but not all, studies revealed that plasma levels of UT II are elevated in patients with HF in correlation with levels of other markers, such as NT-proBNP and ET-1.¹⁷⁰ Second, strong myocardial expression of UT II in end-stage HF correlates with the degree of cardiac impairment.¹⁷⁰ The upregulated UT II in HF may also have a role in the regulation of renal function in HF. In rat models of HF, UT II acted primarily as a renal vasodilator, apparently by an NO–dependent mechanism.¹⁷¹ RPF and GFR, but not urinary Na^+ excretion, were also increased. On the other hand, UT II in control rats led to intense renal vasoconstriction, and a fall in GFR and Na^+ retention.¹⁷¹ In light of the contradictory effects of UT II in different conditions, the clinical application of these data will be challenging.

NEUROPEPTIDES

Because NPY colocalizes and is released with adrenergic neurotransmitters, high circulating NE levels in HF are accompanied by excessive corelease of NPY and plasma levels are correlated with disease severity in HF patients.¹⁷² In contrast, local myocardial levels, like those of NE, were lower than normal in association with decreased Y1 and increased Y2 receptor expression.¹⁷² Because Y1 receptor activation is associated with cardiomyocyte hypertrophy and Y2 receptor activation with angiogenesis, the data in this model suggest that NPY may simultaneously attenuate the maladaptive cardiac remodeling observed in HF and stimulate angiogenesis in the ischemic heart.¹⁷² Similar patterns of receptor expression change were observed in the kidneys, and these were proportional to the degree of renal failure and Na^+ retention.¹⁷² In contrast, administration of NPY in experimental models of HF led to diuresis and natriuresis, probably by increasing ANP release and inhibiting the RAAS.¹⁷³ Therefore in HF, the higher

circulating levels, together with the reduced tissue levels of NPY, could be a counterregulatory mechanism to modulate the vasoconstrictive and Na^+ retaining, as well as the cardiac remodeling, effects of the RAAS and SNS. In addition, the downregulation of Y1 receptors, by reducing vasoconstriction, could contribute to reduced coronary and renal vascular resistance. However, once the stage of decompensated HF is reached, the RAAS and SNS effects likely dominate, thereby overwhelming any favorable effects of NPY.

Levels of other neuropeptides, such as catestatin, may be elevated in HF and have been investigated as potential biomarkers but did not improve diagnostic accuracy over BNP.¹⁷³ In summary, laboratory data on neuropeptides in HF have not translated into clinical application.

APELIN

The expression of apelin and its receptor in the kidney and heart and the involvement of the system in the maintenance of water balance suggested a potential role in HF. Circulating levels rise in early HF but decline in later stages of the disease.^{174,116} However, this decline correlates poorly with severity of HF, making apelin not useful as a biomarker of HF progression.¹⁷⁵

The fact that activation of the apelin receptor induces aquaresis, vasodilation, and a positive inotropic effect suggested the receptor as a potential therapeutic target in HF. Along these lines, acute intravenous injection of apelin to rats with HF following induced myocardial function led to improved systolic and diastolic function. Moreover, more chronic infusions (3 weeks) decreased Ang II–induced cardiac fibrosis and remodeling.¹⁷⁴ In HF patients, acute intravenous apelin increased cardiac output, reduced BP and vascular resistance.¹⁷⁴ No data are yet available on the direct renal effects of apelin in HF, although, by reducing AVP levels and improving the renal microcirculation, apelin might increase aquaresis. In addition, the favorable effects on cardiac function are likely to increase renal perfusion and hence promote diuresis.¹⁷⁶ Stable apelin analogs are currently in development.¹⁷⁷

PEROXISOME PROLIFERATOR–ACTIVATED RECEPTORS

Peroxisome proliferator–activated receptors (PPARs) are nutrient-sensing nuclear transcription factors, of which PPAR γ is of special interest in the context of Na^+ and water retention because of its ligands, the thiazolidinediones (TZDs). TZDs, by increasing insulin sensitivity, are used for the management of type 2 diabetes mellitus. TZDs also decrease circulating free fatty acids and triglycerides, lower BP, reduce levels of inflammatory markers, and reduce atherosclerosis. Moreover, they have a beneficial effect on cardiac remodeling in myocardial ischemia.¹⁷⁸ However, a troubling side effect of TZD is fluid retention mainly resulting from PPAR γ –induced Na^+ reabsorption mediated by increased ENaC expression in collecting duct epithelium.¹⁷⁹ However, TZD may also augment proximal tubular Na^+ reabsorption by upregulation of apical NEH3 , basolateral $\text{Na}^+\text{HCO}_3^-$ cotransporter, and $\text{Na}^+\text{K}^+\text{ATPase}$. These effects are mediated by PPAR γ –induced nongenomic transactivation of the epidermal growth factor receptor and downstream extracellular signal-regulated kinases.¹⁸⁰ Moreover, by reducing SVR, TZD might lead to higher capillary perfusion pressures and fluid extravasation¹⁷⁹;

TZDa are also potent VEGF inducers, leading to increased vascular permeability. In clinical terms, the Na⁺-retaining effect of TZD translates into an increased incidence of HF¹⁷⁹ and therefore they are contraindicated in advanced HF.

Because of the Na⁺-retaining and fluid-retaining effects, as well as other concerns related to increased cardiovascular events on the one hand and favorable effects on the myocardium on the other, the exact role of TZD in HF remains a hotly debated subject.¹⁸¹

In summary, alterations in the efferent limb of volume regulation in HF include both enhanced activities of vasoconstrictor/Na⁺-retaining systems and counterregulatory vasodilatory/natriuretic systems. The magnitude of Na⁺ excretion by the kidneys and therefore the disturbance in volume homeostasis in HF are largely determined by the balance between these antagonistic systems. In the early stages of HF, the vasodilatory/natriuretic systems are important in the maintenance of circulatory and renal function. However, with the progression of HF, the balance shifts toward dysfunction of the vasodilatory/natriuretic systems and enhanced activation of the vasoconstrictor/antinatriuretic systems. The net result is renal circulatory and tubular alterations that result in avid retention of salt and water and edema formation.

RENAL SODIUM RETENTION IN CIRRHOSIS WITH PORTAL HYPERTENSION

Avid Na⁺ and water retention commonly occur in cirrhosis with portal hypertension, leading eventually to ascites, a major cause of morbidity and mortality, with the occurrence of spontaneous bacterial peritonitis, variceal bleeding, and development of hepatorenal syndrome (HRS).^{182,183} As in HF, the pathogenesis of renal Na⁺ and water retention in cirrhosis is related to extrarenal regulation of renal Na⁺ and water handling.

A *sine qua non* for the Na⁺ and water retention in cirrhosis is the development of intrasinusoidal portal hypertension, with values of portal pressure above 12 mm Hg generally being required. In contrast, presinusoidal hypertension alone, as observed in portal vein thrombosis, is not associated with fluid retention. The hallmark of fluid retention in cirrhosis is peripheral arterial vasodilation, in association with renal vasoconstriction. In the early stages of cirrhosis, vasodilation occurs in the splanchnic vascular bed, with arterial pressure maintained through increases in plasma volume and cardiac output, leading to the so-called “hyperdynamic circulation” (overfilling). At this stage, renal Na⁺ and water retention is already evident and aids in the maintenance of EABV.^{6,7,184} However, as cirrhosis progresses, vasodilation in the systemic and pulmonary circulations becomes prominent and cardiac output can no longer compensate for the progressive decrease in SVR.¹⁸⁵ The resulting relative arterial underfilling¹⁸⁴ and reduced EABV leads to unloading of the arterial high-pressure baroreceptors and other volume receptors, in turn, stimulating the classical compensatory neurohumoral response. This response manifests itself as renal, brachial, femoral, and cerebral vasoconstriction and further Na⁺ and fluid retention.¹⁸⁶

Peripheral Arterial Vasodilation

The initial trigger for splanchnic arterial vasodilation is hepatic tissue damage itself, which leads to venous

outflow obstruction, reduced portal venous, blood flow and increased hepatic arterial blood flow. Moreover, the lower the portal venous flow, the higher the hepatic arterial flow (Fig. 13.5A). These changes lead to increased intrahepatic vascular resistance and sinusoidal pressure.¹⁸⁴ Increased hepatic resistance to portal flow causes the gradual development of portal hypertension, collateral vein formation, and shunting of blood to the systemic circulation. As portal hypertension develops, local production of vasodilators—mainly NO, but also carbon monoxide, glucagon, prostacyclin, AM, and endogenous opiates—increases, leading to splanchnic vasodilation.¹⁸⁶ Other contributing factors to splanchnic vasodilation include intestinal bacterial translocation, proinflammatory cytokines, and mesenteric angiogenesis.^{187,188}

The decreases in SVR associated with low arterial BP and high cardiac output account for the well-known clinical manifestations of the hyperdynamic circulation commonly seen in patients with cirrhosis. These include warm extremities, cutaneous vascular spiders, wide pulse pressure, capillary pulsations in the nail bed,¹⁸⁹ and pulmonary vasodilation, associated with the hepatopulmonary syndrome.¹⁹⁰

ABNORMALITIES OF SENSING MECHANISMS IN CIRRHOSIS

Nitric Oxide

The NO system is integrally involved in the pathogenesis of the hyperdynamic circulation and Na⁺ and water retention in cirrhosis, as well as in hepatic encephalopathy, hepatopulmonary syndrome, and cirrhotic cardiomyopathy.¹⁹¹ NO is produced in excess by the vasculature of different animal models of portal hypertension and in cirrhotic patients.¹⁹¹ In animal models, the increased production of NO can be detected at the onset of Na⁺ retention and before the appearance of ascites and NO has been implicated in the impaired vascular responsiveness to vasoconstrictors.¹⁹² Moreover, removal of the vascular endothelial layer abolishes the difference in vascular reactivity between cirrhotic and control vessels.¹⁸⁹

Inhibition of NOS has beneficial effects in experimental models of cirrhosis and in cirrhotic patients. By reducing the high NO production to control levels, the hyperdynamic circulation in cirrhotic rats with ascites was corrected and accompanied by a marked increase in Na⁺ and water excretion and regression of ascites. Concomitant decreases in PRA, aldosterone, and vasopressin concentrations were also observed.^{193,194} In cirrhotic patients, the vascular hyporesponsiveness of the forearm circulation to NE could be reversed by NOS inhibition.¹⁹⁵ Inhibition of NO production also corrected the hypotension and hyperdynamic circulation, led to improved renal function and Na⁺ excretion, and caused a decrease in plasma NE levels in these patients. However, in patients with established ascites, NOS inhibition did not improve renal function.¹⁹⁶

The main enzymatic isoform responsible for the increased systemic vascular NO generation in cirrhosis appears to be eNOS in the systemic and splanchnic circulations.¹⁹¹ Upregulation of eNOS appears, at least in part, to be caused by increased shear stress as a result of portal venous hypertension with increased splanchnic blood flow.¹⁹¹

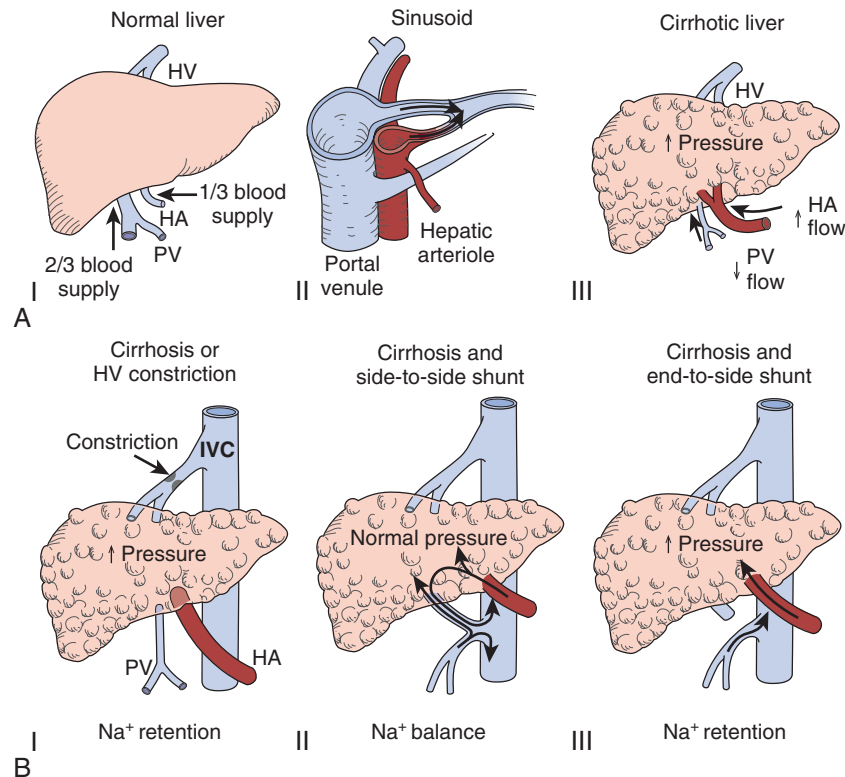


Fig. 13.5 Characteristics of hepatic blood flow. (A) Hepatic circulation. *I*, The normal liver receives two-thirds of its blood flow from the portal vein (PV) and the remaining third from the hepatic artery (HA). *II*, Both the portal venules and hepatic arterioles drain into hepatic sinusoids, but the exact arrangement that allows forward flow of the mixed venous and arterial blood remains unclear. *III*, Cirrhosis increases intrahepatic vascular resistance and sinusoidal pressure. In addition, PV flow is markedly decreased, and HA flow is unchanged or increased. (B) Hepatic vascular hemodynamics and sodium balance. *I*, Cirrhosis or restriction of HV flow increases intrahepatic vascular resistance and sinusoidal pressure, markedly decreasing PV flow and increasing HA flow. Changes in the physical forces or in the composition of the hepatic blood trigger Na^+ retention and edema formation. *II*, Insertion of a side-to-side portocaval shunt decreases sinusoidal pressure and maintains mixing of PV and HA blood, irrigating the liver. Under these conditions and despite cirrhosis, there is no Na^+ retention. *III*, Insertion of an end-to-side portocaval shunt only partially decreases the elevated sinusoidal pressure and prevents mixing of PV and HA blood supplies, inasmuch as the PV blood is diverted to the inferior vena cava (IVC). Under these conditions and, despite normalization of PV pressure, Na^+ retention continues unabated. (Modified from Oliver JA, Verna EC. Afferent mechanisms of sodium retention in cirrhosis and HRS. *Kidney Int.* 2010;77:669–680.)

Increased NO release, as well as eNOS upregulation, in the superior mesenteric arteries was found to precede the development of the hyperdynamic splanchnic circulation.¹⁹¹ In accord with this concept, upregulation of hepatic eNOS (or neuronal NO synthase, nNOS) expression in rats with experimental cirrhosis was associated with a decrease in portal hypertension.¹⁹⁷ However, mice with targeted deletion of eNOS alone or combined deletions of eNOS and inducible NO synthase (iNOS) can still develop hyperdynamic circulation in association with portal hypertension.¹⁹⁴ This suggests that activation of other vasodilatory agents such as PGI_2 , endothelium-derived hyperpolarizing factor, carbon monoxide, and AM may participate in the pathogenesis of hyperdynamic circulation in experimental cirrhosis.¹⁹²

In addition to eNOS, other isoforms may be involved in the generation of the hyperdynamic circulation and fluid retention in experimental cirrhosis. Increased expression of nNOS in mesenteric nerves may compensate partially for eNOS deficiency in eNOS knockout mice and reduce intrahepatic venous resistance and portal hypertension.¹⁹⁷ Also, splanchnic vasodilation is modestly promoted, possibly by modulating neurogenic NE release.¹⁹⁷ In contrast, the role of iNOS remains controversial; some researchers have shown increased iNOS in the superior mesenteric arteries of

animals with experimental biliary cirrhosis but not in other forms of experimental cirrhosis.¹⁹⁷ Specific iNOS inhibition led to peripheral vasoconstriction but had no effect on portal hypertension.¹⁹⁷ iNOS is primarily regulated at the transcription level by many proinflammatory factors, principally nuclear factor- κB (NF- κB), which could be induced by endotoxin, from translocated intestinal bacteria. Interestingly, there is also an interaction between eNOS and iNOS in the vasculature in cirrhosis. Overexpression of eNOS in large arteries results in systemic hypotension and increased blood flow. These effects can be abrogated by activated iNOS in the small splanchnic vessels.¹⁹⁷ Thus overall, available data indicate a predominant role for eNOS deficiency, with possible modulation by both nNOS and iNOS.

In marked contrast to the increased NO generation in the splanchnic and systemic circulation, NO production and endothelial function in the intrahepatic microcirculation are impaired in cirrhotic rats.¹⁹⁷ The resulting paradoxical increase in intrahepatic vascular resistance is likely to result from contraction of myofibroblasts and stellate cells and mechanical distortion of the vasculature by fibrosis.¹⁹⁷ A second mechanism for the increased intrahepatic vascular resistance could be that the locally decreased NO production

shifts the balance in favor of local vasoconstrictors (ET, leukotrienes, thromboxane A₂, Ang II).¹⁹⁸ The increased vascular resistance may also play a role in the pathogenesis of intrahepatic thrombosis and collagen synthesis in cirrhosis.¹⁹⁸

Several cellular mechanisms have been implicated in the upregulation of splanchnic eNOS and downregulation of intrahepatic eNOS. Elevation in shear stress as a result of the hyperdynamic circulation and portal hypertension has already been mentioned and is generally consistent with this well-documented mechanism for upregulating eNOS gene transcription. However, eNOS activity is regulated not only transcriptionally but also posttranscriptionally, by tetrahydrobiopterin (THB₄) and direct phosphorylation of eNOS protein.¹⁹⁷ Furthermore, circulating endotoxins may increase the enzymatic production of THB₄, thereby enhancing mesenteric vascular eNOS activity.¹⁹⁷

Potential contributors to intrahepatic eNOS downregulation include interactions with caveolin, calmodulin, heat shock protein 90, eNOS trafficking inducer,¹⁹⁷ disorders of GC activity,¹⁹⁹ and increased levels of the NO inhibitor, ADMA.¹⁹⁹ In fact, ADMA levels correlate with the severity of portal hypertension during hepatic inflammation and levels are higher in patients with decompensated than compensated cirrhosis.²⁰⁰ Raised ADMA levels have been linked to reduced activity of dimethylarginine dimethylaminohydrolases (DDAHs) that metabolize ADMA to citrulline.²⁰¹ Similarly, patients with alcoholic cirrhosis and superimposed alcoholic hepatitis have higher plasma and tissue levels of ADMA, higher portal venous pressures, and decreased DDAH expression.²⁰⁰ However, attempts at pharmacologic or genetic upregulation of DDAH to reduce ADMA levels and increase NO in experimental cirrhosis have not translated into improved management of decompensated portal hypertension.²⁰²

In the final analysis, the relative importance of the various mechanisms involved in the reduced intrahepatic and increased splanchnic and systemic NOS activity in cirrhosis remains to be determined.

Endocannabinoids

Endogenous cannabinoids are lipid-signaling molecules that mimic the activity of Δ^9 -tetrahydrocannabinol, the main psychotropic constituent of marijuana. *N*-arachidonylethanolamide (anandamide) and 2-arachidonoylglycerol are the most widely studied endocannabinoids that bind the two specific receptors, CB₁ and CB₂. Anandamide also interacts with the vanilloid receptor.²⁰³

In animal models of cirrhosis, both CB₁ and CB₂ receptors and endocannabinoid production are greatly upregulated and anandamide caused a dose-dependent increase in intrahepatic vascular resistance, especially in the isolated perfused cirrhotic liver. The effect appeared to be mediated by CB₁ receptor-enhanced production of COX-derived vasoconstrictive eicosanoids. Also, the CB₁ receptor antagonist, rimonabant, in cirrhotic rats reversed arterial hypotension, increased hepatic vascular resistance, decreased mesenteric arterial blood flow and portal venous pressure, and prevented ascites formation. The reduction in splanchnic blood flow was enhanced by the vanilloid receptor antagonist, capsazepine. These findings indicate that the transient receptor potential vanilloid type 1 protein

and the CB₁ receptor have a dual role in the splanchnic vasodilation characteristic of cirrhosis.²⁰³

Endotoxin was found to be a major stimulus for endocannabinoid generation in monocytes and platelets of cirrhotic animals. This pathway could operate in patients with advanced cirrhosis in whom elevated circulating endotoxin levels are frequently found. Endocannabinoid production could then trigger splanchnic and peripheral vasodilation, arterial hypotension, and intrahepatic vasoconstriction through activation of CB₁ receptors in the vascular wall and perivascular nerves.²⁰³ The potentially favorable effect of CB₁ receptor blockade on Na⁺ excretion opens the possibility of pharmacologic modification of human HRS.

In summary, afferent sensing of volume in cirrhosis is characterized by increased intrahepatic vascular resistance and sinusoidal pressure, decreased portal venous blood flow, and increased hepatic arterial flow. Either changes in intrahepatic physical forces or in the composition of the mixed intrahepatic blood could initiate abnormal Na⁺ retention and edema formation (see Fig. 13.5B). Side-to-side portocaval shunt (currently performed by transjugular intrahepatic portovenous shunt insertion) prevents (if inserted before induction of cirrhosis) or corrects (if inserted after induction of cirrhosis) renal Na⁺ retention. This outcome could result from decreases in sinusoidal pressure or maintenance of the mixing of portal venous and hepatic arterial blood perfusing the liver. In contrast, it diverts blood to the IVC but only partially decreases sinusoidal pressure and prevents mixing of portal venous and arterial hepatic blood supplies. Although portal venous pressure is normalized, Na⁺ retention continues unabated (see Fig. 13.5B). Consequently, end-to-side shunting is no longer used clinically.

Afferent Sensing of Intrahepatic Hypertension

Available data are most consistent with the view that the putative EABV sensor(s) in the hepatic circulation are pathologically activated in cirrhosis, failing to respond to the expanded ECF volume.²⁰⁴ The sensing mechanisms likely respond specifically to elevated hepatic venous pressure with increased hepatic afferent nerve activity. The relays for these impulses consist of two autonomic nerve plexuses, surrounding the hepatic artery and portal vein, respectively.²⁰⁵ These neural networks connect hepatic venous congestion to enhanced renal and cardiopulmonary sympathetic activity.

Occlusion of the IVC at the diaphragm was associated with increases in hepatic, portal, and renal venous pressures and resulted in markedly increased hepatic afferent nerve traffic and renal and cardiopulmonary sympathetic efferent nerve activity. Section of the anterior hepatic nerves eliminated the reflex increase in renal efferent nerve activity²⁰⁵ and hepatic denervation in dogs with IVC constriction increased urinary Na⁺ excretion.²⁰⁶ This effect of hepatic denervation was shown to be mediated by the adenosine A1 receptor in cirrhotic rats.²⁰⁷

Apart from the adenosine-mediated hepatorenal reflex, other currently undefined humoral pathways could provide an anatomic or physiologic basis for the primary effects of alterations in intrahepatic hemodynamics on renal function. Despite the wealth of information on hepatic

volume sensing, the molecular identity, cellular location of the sensor, and what is sensed remain elusive.

Arterial Underfilling

Several mechanisms have been proposed to account for the development of relative hypovolemia. The first is disruption in normal Starling relationships governing fluid movement in the hepatic sinusoids. Unlike other capillaries, these are highly permeable to plasma proteins. As a result, partitioning of ECF between the intravascular (intrasinusoidal) and interstitial (space of Disse) and lymphatic compartments of the liver is determined predominantly by the ΔP along the length of the hepatic sinusoids. Obstruction of hepatic venous outflow promotes enhanced efflux of a protein-rich filtrate into the space of Disse and results in augmented hepatic lymph formation.^{187,208} In parallel, vastly increased hepatic lymph formation is accompanied by increased flow through the thoracic duct.²⁰⁹ When the rate of enhanced hepatic lymph formation exceeds the capacity for return to the intravascular compartment via the thoracic duct, hepatic lymph accumulates as ascites, and the intravascular compartment is further compromised. As liver disease progresses, fibrosis around Kupffer cells lining the sinusoids renders the sinusoids less permeable to serum proteins. Under such circumstances, termed “capillarization of sinusoids,” a decrease in oncotic pressure also promotes transudation of ECF within the hepatic lymph space, as in other vascular beds.²¹⁰

Another consequence of intrahepatic hypertension is transmission of elevated intrasinusoidal pressures to the portal vein. This leads to expansion of the splanchnic venous system, collateral vein formation, and portosystemic shunting, resulting in increased vascular capacitance and diversion of blood flow from the arterial circuit.²¹¹ Both splanchnic and systemic vasodilation occur, and this has been attributed to refractoriness to the vasoconstrictive effects of hormones such as Ang II and catecholamines, although the mechanism remains unknown.²¹² Along with diminished hepatic reticuloendothelial cell function, portosystemic shunting allows various products of intestinal metabolism and absorption to bypass the liver and escape hepatic elimination. Among these products, endotoxins are thought to contribute to perturbations in renal function in cirrhosis, either due to intestinal bacterial translocation, stimulating the release of proinflammatory cytokines (e.g., tumor necrosis factor- α [TNF- α] and interleukin-6), secondary to the hemodynamic consequences of endotoxemia, or through direct renal effects.¹⁸⁷

Levels of conjugated bilirubin and bile acids may become elevated as a result of intrahepatic cholestasis or extrahepatic biliary obstruction. Bile acids directly decrease proximal tubular reabsorption of Na⁺, tending to promote natriuresis.²¹³ This diuretic effect might contribute to the underfilling state in advanced cirrhosis.^{213,214}

Hypoalbuminemia in advanced cirrhosis, either as a result of decreased synthesis by the liver or dilution caused by ECF volume expansion, could also contribute to the development of hypovolemia by diminishing COP in the systemic capillaries and hepatic sinusoids.²¹⁰ In addition, tense ascites might reduce venous return (preload) to the heart, leading to reduced cardiac output and diminished arterial BP.¹⁸⁵

Other factors that may also adversely affect cardiac performance include diminished β -adrenergic receptor signal transduction, cardiomyocyte cellular plasma membrane dysfunction, and increased activity or levels of cardiodepressants, such as cytokines, endocannabinoids, and NO. Although the cardiac dysfunction, termed “cirrhotic cardiomyopathy,” usually is clinically mild or silent, overt HF can be precipitated by stresses such as liver transplantation or transjugular intrahepatic portosystemic shunt (TIPS) insertion.¹⁸⁵ Finally, intravascular volume depletion in cirrhotic patients may be aggravated by vomiting, occult variceal bleeding, and excessive diuretic use, leading to cardiovascular collapse.

Table 13.5 summarizes the various causative factors contributing to underfilling of the circulation in patients with advanced liver disease. In summary, the early stages of compensated cirrhosis are characterized by increased plasma volume, which frequently antedates ascites formation.²¹⁵ However, as cirrhosis progresses, EABV decreases, leading to increased neurohumoral activity (RAAS, SNS, and AVP) and severe Na⁺ and water retention.

Abnormalities of Effector Mechanisms in Cirrhosis

The efferent limb of volume regulation in cirrhosis is similar to that in HF, consisting of adjustments in glomerular hemodynamics and tubular transport mediated by vasoconstrictor/antinatriuretic forces (RAAS, SNS, AVP, and ET) and counterbalanced by vasodilator/natriuretic systems (NPs and PG). Tilting the balance toward Na⁺ retaining forces leads to renal Na⁺ and water retention, as in HF.¹⁸⁶

VASOCONSTRICTOR AND ANTINATRIURETIC (ANTIDIURETIC) SYSTEMS

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

As in HF, the RAAS plays a central role in mediating renal Na⁺ retention in cirrhosis. Although positive Na⁺ balance may already be evident in the preascitic phase of the disease, PRA and aldosterone levels remain within the normal range or may even be depressed at this stage.²¹⁶ With progression of the disease, RAAS activation increases in parallel and aldosterone levels are inversely correlated with renal Na⁺ excretion in preascitic cirrhotic patients, particularly in the upright position.²¹⁶ Moreover, treatment with the ARB,

Table 13.5 Factors Causing Underfilling of Circulation in Cirrhosis

Peripheral vasodilation and blunted vasoconstrictor response to reflex, chemical, and hormonal influences
Arteriovenous shunts, particularly in portal circulation
Increased vascular capacity of portal and systemic circulation
Hypoalbuminemia
Impaired left ventricular function, so-called cirrhotic cardiomyopathy
Diminished venous return secondary to advanced tense ascites
Occult gastrointestinal bleeding from ulcers, gastritis, or varices
Volume losses caused by vomiting and excessive use of diuretics

losartan, at a dosage that did not affect systemic and renal hemodynamics or GFR, was associated with a significant natriuresis,²¹⁷ likely due to inhibition of the local intrarenal RAAS.^{216,217} Indeed, activation of the intrarenal RAAS may precede systemic activation.²¹⁸ In addition, losartan caused a decrease in portal venous pressure in cirrhotic patients with portal hypertension.²¹⁹ The postural-induced RAAS activation and beneficial effects of low-dose losartan treatment in preascitic cirrhosis may be explained by splanchnic venous compartmentalization of the expanded blood volume on standing and translocation toward the central and arterial circulatory beds during recumbency.²¹⁶

In contrast, in Na⁺-retaining cirrhotic patients with ascites, Ang II inhibition, even at low doses, resulted in decreased GFR and Na⁺ excretion.²²⁰ At this stage of the disease, RAAS activation serves to maintain arterial pressure and adequate circulation. Therefore RAAS blockade may lead to a profound decrease in RPP. This scenario might be important in the pathogenesis of the HRS, which is regularly preceded by Na⁺ retention and may be precipitated by a hypovolemic insult. Abnormalities of the renal circulation in HRS include marked diminution of RPF with renal cortical ischemia and increased renal vascular resistance, abnormalities consistent with the known actions of Ang II on the renal microcirculation.²²¹ In this regard, RAAS activation correlates with worsening hepatic hemodynamics and decreased survival in patients with cirrhosis.²²² Therefore ACEIs and ARBs should be avoided in patients with cirrhosis and ascites.

Evolving knowledge on the ACE2, Ang 1–7, Mas receptor pathway has shed new light on the role of the RAAS in the pathogenesis of Na⁺ retention in cirrhosis. In this regard, exogenous Ang 1–7 elicited a marked NO-dependent vasodilatory effect on the Ang-II-evoked vasoconstrictive response in the portal vein of isolated perfused cirrhotic rat liver.^{223,224} The data raise the possibility of reducing intrahepatic resistance and portal pressure by targeted upregulation of the alternate RAAS pathway in the liver.²²⁵

SYMPATHETIC NERVOUS SYSTEM

Activation of the SNS is characteristic of cirrhosis and ascites.²²⁶ Circulating NE levels, as well as urinary excretion of catecholamines and their metabolites, are elevated in patients with cirrhosis and usually are correlated with the severity of the disease. Moreover, high levels of plasma NE in patients with decompensated cirrhosis are predictive of increased mortality.²²⁶ The increased NE levels stem from enhanced SNS activity, rather than reduced dissipation, with nerve terminal spillover from hepatic, cardiac, renal, muscular, and cutaneous innervation.²²⁷ Elevated plasma NE levels were correlated closely with Na⁺ and water retention in cirrhotic patients.²²⁸ In addition, increased efferent renal sympathetic tone, perhaps due to defective arterial and cardiopulmonary baroreflex control, was observed in experimental cirrhosis.^{229,60} This scenario could explain why volume expansion does not suppress enhanced RSNA in cirrhosis.

Concomitant with the increase in NE release, cardiovascular responsiveness to reflex autonomic stimulation may be impaired in patients with cirrhosis.²²⁸ This impairment could be explained partially by increased occupancy of endogenous catecholamine receptors, downregulation

of adrenergic receptors, or a defect in postreceptor signaling.²²⁷ Excessive NO-dependent vasodilation alone could, in fact, account for the vascular hyporesponsiveness in cirrhosis.^{230,231} Also, enhanced release of NPY may be a compensatory mechanism to counteract splanchnic vasodilation by restoring the vasoconstrictor efficacy of endogenous catecholamines.²²⁷

The increase in RSNA and plasma NE levels could contribute to the antinatriuresis of cirrhosis by decreasing total RBF, or its intrarenal distribution, or by acting directly on the tubular epithelium to enhance Na⁺ reabsorption. Patients with compensated cirrhosis may have decreased RBF and, as the disease progresses, RBF tends to decline further, concomitantly with increased sympathetic activity.²²⁶ Indeed, SNS activation in cirrhotic patients is associated with a rightward and downward shift of the RBF-RPP autoregulatory curve such that RBF becomes critically dependent on RPP. This phenomenon was found to contribute to the development of the HRS. Furthermore, insertion of TIPS to reduce portal venous pressure in patients with HRS leads to a fall in plasma NE levels and to an upward shift in the RBF-RPP curve.²³²

Reflex activation of the splenic afferent and renal sympathetic nerves also controls renal microvascular tone. In portal hypertension, the splenorenal reflex-mediated reduction in renal vascular conductance exacerbates Na⁺ and water retention and may eventually contribute to renal dysfunction. Also, increased splenic venous outflow pressure resulting from, but independent of, portal hypertension reflexly activates adrenergic-angiotensinergic and vasodilator mesenteric nerves and the RAAS. Finally, the spleen itself may be the source of a vasoactive factor.^{233,234}

The centrality of SNS overactivity in cirrhosis has been illustrated by the finding that in patients with cirrhosis and increased SNS activity, addition of clonidine or guanfacine to diuretic treatment induces an earlier and enhanced diuretic response, with fewer complications.^{235,236} In advanced cirrhosis, increased SNS activity parallels increases in RAAS and AVP activities.²³⁶ This marked neurohumoral activation probably reflects a shift toward decompensation, characterized by a severe decrease in EABV depletion.²³⁷ Overall, the three pressor systems might be activated by the same mechanisms and operate in concert to counteract the low arterial BP and decrease in EABV.¹⁸⁴

VASOPRESSIN

Impaired water excretion as a result of nonosmotic release of AVP secondary to decreased EABV is frequent in advanced cirrhosis, leading to water retention with hyponatremia.²³⁸ Affected patients also have higher PRA and aldosterone levels and lower urinary Na⁺ excretion.¹⁸⁴ In rats with experimental cirrhosis, plasma levels of AVP were elevated in association with overexpression of hypothalamic AVP mRNA and diminished pituitary AVP content.²³⁹ Concomitantly, increased expression of AQP2, the AVP-regulated water channel in the collecting duct, was significantly diminished by the AVP receptor antagonist, terlipressin, indicating the important role of AQP2 in the water retention associated with hepatic cirrhosis.²⁴⁰

As noted earlier, AVP supports arterial BP through its action on the V₁ receptors on vascular smooth muscle cells, whereas the V₂ receptor is responsible for water transport in

the collecting duct.²⁴¹ The availability of selective blockers of these receptors has provided clear evidence for the dual roles of AVP in pathogenesis of cirrhosis.^{105,241} Thus administration of a V_2 receptor antagonist to cirrhotic patients, as well as to rats with experimental cirrhosis, increases urine volume, decreases urine osmolality, and corrects hyponatremia^{105,241} (see further discussion in “Specific Treatments Based on the Pathophysiology of Sodium Retention in Cirrhosis” later).

AVP also increases the synthesis of the vasodilatory PGE_2 and PGI_2 in renal and other vascular beds, as well as in the collecting duct. This increase may offset the vasoconstrictive and hydroosmotic effects of AVP in cirrhosis.²⁴²

ENDOTHELIN

Levels of ET-1 and big ET-1 in plasma, splanchnic, and renal venous beds are markedly elevated in patients with cirrhosis and ascites, as well as in the HRS.^{243,244} Levels correlate positively with portal venous pressure and cardiac output and inversely with central blood volume.²⁴⁴ The rise in ET-1 is accompanied by a reduction in ET-3 levels, and the consequently elevated ET-1/ET-3 ratio is associated with a poor outcome of portal hypertension.²⁴⁵ In animal models of cirrhosis with portal hypertension, ET-A receptor activation and attenuated ET-B receptor repression on the portal vein have been reported.²⁴⁶ ET-B receptor blockade led to sinusoidal constriction and hepatotoxicity,²⁴⁷ whereas the dual ET-A and B receptor blocker, tezocentan, had no effect on hepatic blood flow.²⁴⁸

In humans with HRS, ET-1 and big ET-1 levels were significantly reduced in portal and renal veins 1 to 2 months after TIPS insertion, with a parallel increase in creatinine clearance and urinary Na^+ excretion.²⁴³ Similar improvements have been observed within 1 week after successful orthotopic liver transplantation.²⁴⁹ Conversely, temporary occlusion of TIPS by angioplasty balloon inflation led to a transient increase in portal venous pressure, increased plasma ET-1, marked reduction of RPF, and increased intrarenal generation of ET-1.²⁵⁰

The importance of the intrarenal ET system has been demonstrated in a rat model of HRS.²⁵¹ Plasma ET-1 increased twofold after the onset of liver and renal failure, and the ET-A receptor was upregulated in the renal cortex. Bosentan, a nonselective ET receptor antagonist, prevented the development of renal failure when given before or 24 hours after onset of liver injury.²⁵¹

Increased intrahepatic production of ET probably also contributes to the development of portal (and pulmonary) hypertension in cirrhosis through contraction of the stellate cells and a concomitant decrease in sinusoidal blood flow.²⁵² To summarize, the hemodynamic changes in cirrhosis with refractory ascites could be related to local ET-1 production by the splanchnic and renal vascular beds. After TIPS and orthotopic liver transplantation, there are improvements in both ET and other vasoconstrictive factors (e.g., RAAS and vasopressin).²⁵³ Therefore the contribution of the intrarenal ET system relative to other vasoconstrictor hormones in the pathogenesis of the HRS remains speculative.

APELIN

The possible involvement of apelin in the pathogenesis of cirrhosis was suggested by the raised plasma levels²⁵⁴ and

enhanced expression of its receptor in proliferated arterial capillaries directly connected with sinusoids.^{255,256} In addition, an apelin receptor antagonist led to a reduction in the raised cardiac index, reversal of the increased total peripheral resistance, and improvement in Na^+ and water excretion in rats with experimental cirrhosis.²⁵⁴ However, to date no therapeutic role of apelin antagonism in the management of severe HRS has been elucidated, possibly owing to the complex effects of apelin on glomerular hemodynamics.²⁵⁷

VASODILATORS/NATRIURETICS

Apart from their role in the hyperdynamic circulation characteristic of advanced cirrhosis, vasodilators play an important part in the pathogenesis of renal Na^+ retention. The principal vasodilators involved are NPs and PGs.

NATRIURETIC PEPTIDES

Atrial Natriuretic Peptide

In recent years, measurements of BNP and NT-proBNP have largely superseded ANP as a biomarker of cirrhosis and portal hypertension. Nevertheless, the role of NPs in the pathogenesis of HRS was largely elucidated through studies on ANP, and these are summarized here. Plasma ANP is elevated in cirrhosis at all stages, irrespective of the EABV.^{258,259} In the preascitic stage, increased plasma ANP may be important for the maintenance of Na^+ homeostasis, but with progression of the disease, resistance to the natriuretic action of the peptide develops.^{258,259} The high levels of ANP mostly reflect increased cardiac release rather than impaired clearance.²⁶⁰ The stimulus for increased cardiac ANP synthesis and release in early cirrhosis is likely increased left atrial size caused by overfilling of the circulation, secondary to intrahepatic hypertension-related renal Na^+ retention.²⁶¹

In addition to elevated ANP, preascitic patients also had significantly elevated left and right pulmonary volumes, despite normal BP, PRA, aldosterone, and NE levels.²⁶² High Na^+ intake in these patients resulted in weight gain and positive Na^+ balance for 3 weeks, followed by a return to normal Na^+ balance, thereby preventing fluid retention and the development of ascites.²⁶³ The factors responsible for maintaining relatively high levels of ANP during the later stages of cirrhosis, in association with arterial underfilling, may be related to a futile cycle of mutual interactions between vasoconstrictor/ Na^+ -retaining and vasodilatory/natriuretic forces. The fact that ANP levels do not increase further as patients proceed from early to late decompensated stages of cirrhosis would be consistent with this explanation. Furthermore, infusion of Ang II mimicked the nonresponder state by causing patients with cirrhosis, who still responded to ANP, to become unresponsive.²⁶⁴ This Ang II effect occurred at proximal (decreased distal delivery of Na^+) and distal nephron sites to abrogate ANP-induced natriuresis and was reversible. The importance of distal solute delivery was confirmed using mannitol, which also resulted in an improved natriuretic response to ANP in responders but not nonresponders.^{265,266} ANP resistance was also ameliorated by endopeptidase inhibitors, renal

sympathetic denervation, peritoneovenous shunting, and orthotopic liver transplantation.²⁶⁷⁻²⁷⁰

To summarize, ANP resistance is best explained by an effect of decreased delivery of Na^+ to ANP-responsive distal nephron sites (glomerulotubular imbalance caused by abnormal systemic hemodynamics and activation of the RAAS) combined with an overriding effect of more powerful antinatriuretic factors to overcome the natriuretic action of ANP in the medullary collecting tubule.²⁷¹ The latter effect could result from decreased delivery, as well as permissive paracrine/autocrine cofactors, such as PGs and kinins.

Brain Natriuretic Peptide and C-Type Natriuretic Peptide

BNP levels are also elevated in cirrhosis with ascites and, like ANP, its natriuretic effect is blunted in these patients.²⁷²⁻²⁷⁵ Plasma BNP levels may be correlated with cardiac dysfunction²⁷³ and severity of disease and may be of prognostic value in the progression of cirrhosis.^{275,276} Plasma CNP levels in cirrhotic preascitic patients, although normal, were directly correlated with natriuresis and urine volume²⁷⁷ and inversely correlated with arterial compliance but not SVR.²⁷⁸ These data suggested that compensatory downregulation of CNP occurs in cirrhosis when vasodilation persists and that regulation of large and small arteries by CNP may differ.

In contrast with the preascitic stage, patients with more advanced disease and impaired renal function had lower plasma and higher urinary CNP levels than those with intact renal function. Moreover, urinary CNP was correlated inversely with urinary Na^+ excretion. In patients with refractory ascites or HRS treated with terlipressin infusion or TIPS (see “Specific Treatments Based on the Pathophysiology of Sodium Retention in Cirrhosis” later), urinary CNP declined and urinary Na^+ excretion increased 1 week later.^{277,279} Thus CNP may have a significant role in renal Na^+ handling in cirrhosis.

Finally, *Dendroaspis* NP levels were found to be increased in cirrhotic patients with ascites but not in those without, and levels were correlated with disease severity.²⁸⁰ The significance of these findings remains unknown.

PROSTAGLANDINS

As noted, PGs modulate the hydroosmotic effect of AVP and protect RPF and GFR when the activity of endogenous vasoconstrictor systems is increased. These properties of PG appear to be critical in decompensated cirrhosis with ascites but no renal failure. Such patients excrete more vasodilatory PG than healthy subjects, suggesting that renal production of PG is increased.²⁸¹ Similarly, in experimental cirrhosis, there is increased synthesis and activity of renal and vascular PG, as well as upregulation of COX-2.^{281,282} PGE₂ upregulation in the thick ascending limb of preascitic cirrhotic rats is mediated by downregulation of calcium-sensing receptors (CaSRs). This maneuver resulted in increased expression of the NKCC2, increased Na^+ reabsorption in this segment, and augmented free water reabsorption in the collecting duct. The effects were reversed by the CaSR agonist poly-L-arginine.²⁸³

Not surprisingly, nonselective COX inhibitors resulted in a significant decrease in GFR and RPF in cirrhotic patients, with or without ascites. The decrement in renal hemodynamics varied directly with the degree of Na^+

retention and neurohumoral activation, so patients with high PRA and NE levels were particularly sensitive to these adverse effects.^{281,282} These negative effects of COX inhibition appear to be solely COX-1 dependent because selective COX-2 antagonists spare renal function in both human and experimental cirrhosis, even with ascites.^{281,282} The favorable renal effect of selective COX-2 antagonists may be indirect and related to hepatic upregulation of COX-2 in that celecoxib can ameliorate portal hypertension by hepatic antiangiogenic and antifibrotic actions.²⁸⁴

In contrast with nonazotemic patients with cirrhosis and ascites, patients with HRS have reduced renal synthesis of vasodilatory PG.²⁸⁵ However, treatment with intravenous PGE₂ or its oral analog, misoprostol, did not improve renal function in HRS patients.²⁸⁶ This PGE₂ resistance may be related to overwhelming neurohumoral vasoconstrictive/antinatriuretic effects and may be crucial in the pathogenesis of HRS.²⁸¹

INTEGRATED VIEW OF THE PATHOGENESIS OF SODIUM RETENTION IN CIRRHOSIS

Portal hypertension leads to increased intestinal permeability, bacterial translocation, endotoxemia, and exposure to bacterial DNA. In turn, hepatic NO and PG increase, leading to splanchnic, then systemic vasodilation, with increased cardiac output, decreased SVR, and arterial pressure (“hyperdynamic circulation”). The resulting imbalance between vascular capacity and plasma volume induces baroreceptor and subsequent neurohumoral activation (RAAS, SNS) leading to systemic and renal vasoconstriction, Na^+ retention, and maintained circulation (compensated cirrhosis). In addition, a yet elusive hepatic-derived factor may also act directly to induce renal Na^+ retention (overflow theory). If there is enough compensatory increase in systemic natriuretic factors and renal synthesis of PG and NO, renal function is also maintained. However, as cirrhosis advances and portal hypertension worsens, features of the HRS appear: EABV falls (underfilling), vasoconstrictive/antinatriuretic factors dominate over vasodilatory/natriuretic factors, GFR declines, and Na^+ retention is further exacerbated, leading to edema and ascites. Also, at an advanced stage, nonosmotic release of AVP becomes prominent, leading to impaired water excretion and hyponatremia.

CLINICAL MANIFESTATIONS OF HYPERVOLEMIA

Apart from the clinical manifestations of the underlying disease, the symptoms and signs of hypervolemia per se depend on the amount and relative distribution of fluid between the intravascular and interstitial spaces. Arterial volume overload is manifested as hypertension, whereas venous overload is characterized by raised jugular venous pressure (JVP). Interstitial fluid accumulation appears as peripheral edema, effusions in the pleural or peritoneal cavity (ascites) or in the alveolar space (pulmonary edema), or a combination of these. If cardiac and hepatic functions are normal and transcapillary Starling forces intact, the excess volume is distributed proportionately throughout the ECF compartments. In this situation, the signs of hypervolemia will be hypertension and raised JVP. Peripheral edema appears only when interstitial volume overload exceeds 3 L, usually due to ongoing renal Na^+ retention.

When cardiac systolic function is impaired, as a result of myocardial, valvular, or pericardial disease, pulmonary and systemic venous hypertension predominate and systemic BP may be low as a result of disproportionate fluid accumulation in the venous rather than the arterial circulation. Disruption in transcapillary Starling forces, as found in advanced cardiac and hepatic disease, may lead to fluid transudation into the pleural and peritoneal spaces, manifested as pleural effusions and ascites, respectively.

As already mentioned, advanced cirrhosis or fulminant hepatic failure leads to ascites and oliguric renal failure in the absence of diuretics, nephrotoxins, shock, and without evidence of significant intrarenal pathology. This is referred to as HRS.^{287,288} Two subtypes of HRS have been defined. Type 1 is characterized by a rapid decline in renal function (doubling of serum creatinine level to >2.5 mg/dL, or 50% reduction in creatinine clearance to <20 mL/min) over a 2-week period. Typically, an acute precipitating factor can be identified. Type 2 develops spontaneously and progressively over months (serum creatinine level >1.5 mg/dL, or creatinine clearance <40 mL/min). To redefine HRS type 1, the International Club of Ascites has recently adopted the KDIGO criteria for AKI and coined the term HRS-AKI. The diagnostic criteria include 1. a rise in serum creatinine of 0.3 mg/dL and/or 50% or more from baseline that is known, or presumed, to have occurred within the prior 7 days; and 2. no response after 2 consecutive days of diuretic withdrawal and plasma volume expansion with albumin (1 g/kg of body weight, with a maximum of 100 g/day).^{287,288} This more sensitive definition allows the possibility for earlier treatment of HRS but has yet to be tested in terms of prognostic significance.^{184,237} Furthermore, in decompensated liver cirrhosis, ascites and peripheral edema are essential for guiding therapy. For example, grade 1 mild ascites, detected by ultrasound only, usually responds to Na⁺ restriction and, if needed, diuretics (see later), whereas grade 3 large ascites with marked abdominal distension may not respond to medical therapy and requires large-volume paracentesis combined with albumin as the initial treatment. Also, when present, peripheral edema enables clinicians to reduce the patient's body weight up to 1 kg/day but with greater caution in its absence (<0.5 kg/day).²⁸⁹ HRS is further discussed in [Chapter 28](#).

DIAGNOSIS

The diagnosis of hypervolemia is usually evident from the clinical history and physical examination. Any combination of peripheral edema, raised JVP, pulmonary crepitations, and pleural effusions is likely to be diagnostic. In the presence of these findings, high BP would suggest renal failure as the cause of hypervolemia, whereas relatively low BP would point to severe (stage D) HF or advanced cirrhosis as the precipitating factor. In more enigmatic cases, in which dyspnea is the sole complaint and clinical findings are minimal, measurement of plasma biomarkers such as BNP, NT-proBNP, and MR-proANP may help distinguish between cardiac and pulmonary causes of the dyspnea, as well as for risk stratification.^{290,291,292} Imaging studies are also recommended including chest x-ray (CXR) and a four-zone lung ultrasound (LUS) for the assessment of B-lines, along with transthoracic echocardiography.^{290,293}

Interestingly, in one prospective observational study, poor association was found between patient-reported symptoms and markers of heart congestion. Among 322 patients hospitalized for acute HF, subjective scoring, using the Kansas City Cardiomyopathy Questionnaire total symptom score (KCCQ-TSS), severity correlated only with peripheral edema but not LUS, NT-proBNP, CXR, jugular distension, or lung crackles. The authors suggested that marked reduction in KCCQ-TSS in patients with acute HF may not be solely explained by pulmonary congestion.²⁹³

Simple laboratory tests may aid in confirming the clinical diagnosis. An elevated cardiac troponin level is consistent with myocardial damage, and high levels may be observed in acute decompensated HF.^{294,295} Transaminase levels may be raised in hepatic disease and hypoalbuminemia would be consistent with hepatic cirrhosis or nephrotic-range proteinuria caused by glomerular disease. The latter, of course, is confirmed by appropriate urine testing.

When BP is low, evidence of prerenal azotemia (increased ratio of blood urea nitrogen [BUN] to creatinine) may be found, as in advanced cardiac or hepatic failure (cardiorenal syndrome and HRS, respectively); intrinsic renal failure—proportionate increases in BUN and creatinine—may also ensue (see [Chapter 28](#) for detailed discussion). Low EABV in the presence of hypervolemia is confirmed by a low urine Na⁺ concentration or low fractional excretion of Na⁺, indicative of secondary renal Na⁺ retention.

TREATMENT

Therapy for volume overload can be divided into management of the volume overload itself and prevention or minimization of its recurrence and associated morbidity and mortality. The critical first step is recognition and treatment of the underlying cause of hypervolemia. Thus when the EABV is significantly reduced, as in cardiac and hepatic failure (or severe nephrotic syndrome), hemodynamic parameters should be optimized. Otherwise, therapy to induce negative Na⁺ balance is associated with an enhanced risk for worsening hemodynamic compromise.

Once the EABV is restored, negative Na⁺ balance can be induced by dietary Na⁺ restriction, diuretics, and extracorporeal ultrafiltration. The degree of hypervolemia and clinical urgency for Na⁺ removal determine which modality should be used. Therefore in a patient with life-threatening pulmonary edema, immediate intravenous loop diuretics are indicated; if high doses of these drugs do not induce significant diuresis, then extracorporeal ultrafiltration may be lifesaving. At the other extreme, a hypertensive patient with mild volume overload and preserved renal function may require only dietary salt restriction and a thiazide diuretic.

Once acute hypervolemia has been controlled, therapy is directed toward the prevention or minimization of further acute episodes and improvement in prognosis. In addition to maintenance diuretic treatment, several strategies, based on the pathophysiologic process of Na⁺ retention, are available clinically or are under experimental development.

SODIUM RESTRICTION

Until recently, the prevailing belief was that effective management of hypervolemia of any cause must include Na⁺

restriction. Without this intervention, the success of diuretic therapy was thought to be limited because the relative hypovolemia induced by diuretics led to compensatory Na^+ retention, increased diuretic dosage, further reduction in EABV, and still more renal Na^+ retention. However, this view was challenged by a randomized controlled trial in patients with acute decompensated HF. In this study, no difference in weight loss or clinical stability was observed at 3 days between the group restricted to 800 mg sodium/day and those with a more liberal sodium intake. Moreover, those on the severely sodium-restricted diet were significantly thirstier than their less-restricted counterparts.²⁹⁶ The SODIUM-HF study also showed that the reduction of dietary sodium to <100 mmol in ambulatory HF patients did not reduce the primary composite outcome of cardiovascular-related admission to hospital, cardiovascular-related emergency department visit, or all-cause death within 12 months in the intention-to-treat (ITT) population. However, there are still evidence gaps and future large studies are needed to evaluate the efficacy and safety of specific dietary interventions, sodium restriction, and fluid restriction for the prevention and treatment of HF.^{290,297} In light of the current debate, a reasonable goal is to restrict Na^+ intake to about 2 to 3 g of Na^+ /day.^{290,298-301} Because of the generally poor palatability of salt-restricted diets, salt substitutes may be used; however, these preparations usually contain high concentrations of potassium and must be used with caution by patients with renal impairment or those taking potassium-retaining RAAS antagonists.

In hospitalized patients, extra attention must be paid to amounts and types of intravenous fluids administered. A frequent scenario encountered by nephrologists consulted in internal medicine departments is a patient receiving intravenous saline together with high-dose diuretics. The usual rationale offered for this combination is that the saline will expand the intravascular volume while the diuretic will mobilize excess interstitial volume. This logic has no sound physiologic or therapeutic basis because both modalities operate principally in the intravascular space. Moreover, the combination of saline and furosemide in HF is associated with adverse outcomes.³⁰² Furthermore, water restriction is also inappropriate, except in the presence of accompanying hyponatremia (plasma $\text{Na}^+ <135$ mmol/L). The same Na^+ restriction applies to patients with cirrhosis.²⁸⁹

DIURETICS

Diuretics are classified according to their sites of action along the nephron and are discussed in detail in [Chapter 49](#). The reader is also referred to comprehensive reviews.³⁰³⁻³⁰⁶ Diuretics are described briefly here in relation to the treatment of hypervolemia.

Proximal Tubule Diuretics

The prototype proximal tubular diuretic is acetazolamide, a carbonic anhydrase inhibitor that inhibits the reabsorption of sodium bicarbonate. Prolonged use may cause hyperchloremic metabolic acidosis, and the drug is usually used in the management of chronic glaucoma rather than for reducing volume overload. Another proximally acting diuretic is the thiazide-like metolazone, which also inhibits the NaCl cotransporter in the distal tubule. The proximal action of metolazone may be associated with phosphate

loss greater than that seen with traditional thiazides.³⁰⁷ In general, metolazone is used as an adjunct to loop diuretics in resistant HF.³⁰⁸ Mannitol, which also inhibits proximal tubular reabsorption,³⁰⁹ can be used in combination with furosemide for the management of acute decompensated HF.³¹⁰

Loop Diuretics

This group includes the most powerful diuretics, furosemide, bumetanide, torsemide, and ethacrynic acid. They act by inhibiting transport via the NKCC2 in the apical membrane of the thick ascending limb of the loop of Henle (see [Chapter 6](#)).³¹¹ They are used for the treatment of severe hypervolemia and hypertension, especially in advanced CKD. In the TRANSFORM-HF trial, among patients discharged after hospitalization due to HF, no statistically significant differences in all-cause mortality or rehospitalizations were observed between patients randomized to torsemide or furosemide over 1 year.³¹² The following TRANSFORM-Mechanism study showed that enhanced natriuresis from higher diuretic doses in the torsemide group was offset by greater neurohormonal activation and kidney dysfunction.³¹³ In the Pragmatic Urinary Sodium-based algorithm in Acute Heart Failure (PUSH-AHF) study, natriuresis-guided diuretic therapy (i.e., prompting treatment intensification when spot urinary Na^+ levels were <70 mmol/L and/or diuresis <150 mL/hour, in bumetanide-treated subjects) was shown to improve 24-hour natriuresis, compared with standard of care (409 ± 178 mmol compared with 345 ± 202 mmol, respectively; $P = 0.0061$).³¹⁴ However, no significant differences between the two arms were observed for the combined primary endpoint of time to all-cause mortality or first HF rehospitalization at 180 days (31% in each group; $P = 0.6980$). Likewise, the duration of the index hospitalization was comparable between the two groups.³¹⁴ Two newer studies used a combination of diuretics in hospitalized patients with decompensated HF.^{315,316} A total of 519 patients underwent randomization in the Acetazolamide in Decompensated Heart Failure with Volume Overload (ADVOR) trial to receive either intravenous acetazolamide (500 mg once daily for 3 days) or placebo, added to standardized intravenous loop diuretics, and were followed for a mean of 3 months.^{315,317} Successful decongestion occurred more often in the acetazolamide group than in the placebo group (42.2% vs. 30.5%, respectively; risk ratio, 1.46 (95% CI, 1.17–1.82); $P < 0.001$) and was associated with a shorter stay in the hospital and a similar safety profile. Mortality and rehospitalization were not different between groups. Of note, this trial was conducted in European countries, limiting therefore the racial and ethnic diversity of the participants. In future trials, it will be interesting to examine the relative benefits and risks of acetazolamide treatment as opposed to other adjunctive diuretics and SGLT2 inhibitors.

The addition of oral hydrochlorothiazide (HCTZ) or placebo to intravenous furosemide was tested in the CLOROTIC trial.³¹⁶ HCTZ was given orally within the first day of hospitalization for acute decompensated HF. HCTZ dose was adjusted to eGFR (eGFR >50 mL/min: 25 mg daily; 20–50 mL/min: 50 mg daily; and <20 mL/min: 100 mg daily). The study was terminated due to slow enrollment (6914 subjects were screened and 230 were randomized).

The treatment combination of furosemide with HCTZ led to increased diuretic response and weight loss (adjusted estimated difference -1.14 kg; $P = 0.002$) without significant change in patient-reported dyspnea. After randomization, increase in serum creatinine levels >26.5 $\mu\text{mol/L}$ or a decrease in serum eGFR $>50\%$ compared with baseline levels were recorded in 46.5% in the HCTZ group compared with only 17.2% in the placebo group ($P < 0.001$). There was no difference between the two arms in 30- or 90-day mortality, hypokalemia ≤ 2.5 mmol/L, and hyponatremia ≤ 125 mmol/L.³¹⁶

Loop diuretics may lead to hypokalemia, intravascular volume depletion, and worsening prerenal azotemia, especially in older patients and those with reduced EABV. They are also hypercalciuric.³¹⁸

Distal Tubule Diuretics

Diuretics in this segment block the apical NaCl cotransporter and include hydrochlorothiazide, chlorthalidone, and metolazone (see also “Proximal Tubule Diuretics” earlier). They are typically used as adjuncts to loop diuretics in resistant HF, particularly metolazone. Inhibition of Na⁺ reabsorption by diuretics in the proximal tubule (except for carbonic anhydrase inhibitors), loop of Henle, and distal tubule leads to increased solute delivery to the collecting duct. Consequently, potassium and proton secretion are enhanced, which may lead to hypokalemia and metabolic alkalosis.³¹⁹ Thiazides are also hypocalciuric.³¹⁸

Collecting Duct Diuretics

Collecting duct (K⁺-sparing) diuretics operate by competing with aldosterone for occupation of the mineralocorticoid receptor (spironolactone, eplerenone) or by direct inhibition of the ENaC (amiloride and triamterene).³¹¹ As their alternative name implies, important side effects of this group are hyperkalemia and metabolic acidosis, resulting from concomitant suppression of K⁺ and proton secretion. Therefore they are widely used in combination with thiazide and loop diuretics to minimize hypokalemia. Aldosterone antagonists are especially useful in the management of disorders characterized by secondary hyperaldosteronism, such as cirrhosis with ascites.^{289,320} Moreover, these drugs are cardioprotective and renoprotective via nonepithelial mineralocorticoid receptor blockade (see “Pathophysiology” and “Specific Treatments Based on the Pathophysiology of Heart Failure” in this chapter; also see Chapter 12).

Diuretic Resistance

As noted, when Na⁺ retention is severe and resistant to conventional doses of loop diuretics, combinations of diuretics acting at different nephron sites may produce effective natriuresis. Another method for overcoming diuretic resistance is the administration of a bolus dose of loop diuretic to yield a high plasma level, followed by high-dose continuous infusion. Alternately, high doses given intermittently may be successful in reversing diuretic resistance. There appears to be no difference between continuous infusion and repeated bolus administration of high or low doses of furosemide in improvement of shortness of breath or volume of diuresis at 72 hours after admission for acute decompensated HF.^{305,321}

Whichever method is used to treat diuretic-resistant hypervolemia, plasma Na⁺, K⁺, Mg²⁺, Ca²⁺, phosphate, BUN, and creatinine should be monitored carefully, and any deviations appropriately corrected. Rarer side effects of diuretics include cutaneous allergic reactions, acute interstitial nephritis, pancreatitis, and, rarely, blood dyscrasias.³²²

EXTRACORPREAL ULTRAFILTRATION

On occasion, extreme resistance to diuretics occurs, often accompanied by renal functional impairment. In such cases, effective removal of volume excess may be achieved by ultrafiltration using hemofiltration, hemodialysis, peritoneal dialysis (see Chapters 62–64), or small devices designed for isolated ultrafiltration (UF).^{321,323} Chronic ambulatory peritoneal dialysis may also reduce hospitalization rates in patients with resistant HF who are not candidates for surgical intervention.³²⁴ However, a randomized controlled trial comparing intensive diuretic therapy with UF for the management of acute decompensated HF with worsened renal function (CARRESS-HF) was halted early because of lack of benefit of UF and an excess of early and late (60 days) adverse events.³²⁵ Moreover, UF was inferior to diuretic therapy in terms of the bivariate primary endpoint of body weight and rise in serum creatinine level at 96 hours after commencement of therapy. Importantly, a Cochrane systematic review, based on RCTs comparing UF with pharmacotherapies in acute HF, indicated an uncertain effect of UF on all-cause mortality at 30 days or less due to overall small sample size.³²⁶ At the longest available follow-up, UF may have little to no effect on all-cause mortality.^{323,325–328} Current guidelines advocate to conduct further investigation related to various aspects of UF including patient selection, fluid removal rates, venous access, prevention of therapy-related complications, and cost.²⁹⁰ Therefore UF should currently be reserved for those patients with unequivocal diuretic resistance and as part of an algorithmic stepped approach to HF and related AKI management.^{321,329,330}

SPECIFIC TREATMENTS BASED ON THE PATHOPHYSIOLOGY OF HEART FAILURE

Because the clinical situation of an HF patient at any given time depends on the delicate balance between vasoconstrictor/antinatriuretic and vasodilator/natriuretic factors, any treatment that can tip the balance in favor of the latter should be efficacious.^{290,297,330,331} Thus treatment is aimed at either pharmacologically increasing natriuretic or reducing antinatriuretic mechanisms. The principal approaches include reducing RAAS or SNS activity or increasing NP activity.

INHIBITION OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

On the basis of maladaptive actions of locally produced or circulatory Ang II, numerous studies have shown that ACE inhibition and ARBs improve renal function, cardiac performance, and life expectancy of HF patients.^{75,332} A small decline in GFR is occasionally observed as a result of

blockade of Ang II–induced preferential efferent arteriolar constriction, which leads to a sharp fall in glomerular capillary pressure, but this is usually not clinically significant. Because patients with HF cannot overcome the Na⁺-retaining action of aldosterone, blockade of the latter by spironolactone or eplerenone induces substantial natriuresis in these patients.³³²

Overall, the effect of RAAS inhibition on renal function in HF depends on a multiplicity of interacting factors. On the one hand, RBF may improve as a result of lower efferent arteriolar resistance. Systemic vasodilation may be associated with a rise in cardiac output. Under such circumstances, reversal of the hemodynamically mediated effects of Ang II on Na⁺ reabsorption would promote natriuresis. Moreover, RAAS inhibition could facilitate the action of NPs to improve GFR and enhance Na⁺ excretion. On the other hand, Ang II–induced elevation of the single-nephron filtration fraction facilitates preservation of GFR in the presence of diminished RPF. In patients with precarious renal hemodynamics, a fall in systemic arterial pressure below the autoregulatory range, combined with removal of the Ang II effect on glomerular hemodynamics, may cause severe deterioration of renal function. The net result depends on the integrated sum of these physiologic effects, which, in turn, depends on the severity of HF (Table 13.6).

In addition to promoting Na⁺ retention, the RAAS contributes to vascular and cardiac remodeling by inducing perivascular and interstitial fibrosis in HF.^{75,333} In accordance with this mechanism, the addition of small doses of aldosterone inhibitors to standard therapy, including ACEIs or ARBs, substantially reduces mortality rate and morbidity in HF patients.^{75,332} In contrast, combinations of ACEI and ARB are not useful and, often, are detrimental, owing to increased risk of hypotension, AKI, and hyperkalemia.³³³ However, the combination of RAAS inhibitor, aldosterone inhibitor β -blocker, and low-dose loop diuretics is associated with a better prognosis than chronic use of increasing loop-diuretic dose alone.³³⁴

Finally, all patients receiving one or more RAAS inhibitors require careful follow-up for detection and management of hyperkalemia, particularly those with renal dysfunction.^{297,333}

β -BLOCKADE

Insofar as β -blockade is now the standard of care in the management of HF, this review would not be complete without mention of these drugs. However, because their effect on HF is not directly related to Na⁺ and water, this important therapy is not elaborated further in this chapter. The reader is referred to this meta-analysis for up-to-date information.³³⁵

NITRIC OXIDE DONOR AND REACTIVE OXYGEN SPECIES/PEROXYNITRATE SCAVENGERS

Because NO signaling is disrupted in HF, achieving NO balance by NO donors or selective NOS inhibitors could be important for correcting the pathophysiologic process of HF.³³⁶ In this regard, the beneficial effects of combined isosorbide dinitrate (NO donor) and hydralazine (reactive oxygen species and peroxynitrite scavenger)

Table 13.6 Renal Effects of Renin-Angiotensin-Aldosterone System Inhibition in Heart Failure

Factors Favoring Improvement in Renal Function

Maintenance of Na⁺ balance
Reduction in diuretic dosage
Increase in Na⁺ intake
Mean arterial pressure >80 mm Hg
Minimal neurohumoral activation
Intact counterregulatory mechanisms

Factors Favoring Deterioration in Renal Function

Evidence of Na⁺ depletion or poor renal perfusion
Large doses of diuretics
Increased urea/creatinine ratio
Mean arterial pressure <80 mm Hg
Evidence of maximal neurohumoral activation
AVP-induced hyponatremia
Interruption of counterregulatory mechanisms
Coadministration of prostaglandin inhibitors
Adrenergic dysfunction (e.g., diabetes mellitus)

AVP, Arginine vasopressin.

therapy, particularly in African-American patients, are well established.³³⁷ Newer methods for enhancing NO activity include NOS and GC stimulators and inhibitors of GC breakdown.^{297,331,336,338} Relaxin-mediated and Waon-mediated NOS stimulation, in particular, are currently being tested in a phase 3 clinical trial. Relaxin mediates NO production both immediately via stimulation of eNOS and chronically via increased expression of iNOS. Treatment of acute HF with serelaxin, recombinant relaxin-2, was associated with dyspnea relief and reduced 180-day mortality, but had no effect on hospital readmission rates. Serelaxin treatment was well tolerated and safe.³³⁹

Waon, or “soothing warm” therapy, which causes vasodilation via eNOS stimulation, is a thermal treatment developed in Japan and shown in a clinical trial to improve NYHA class and exercise tolerance in chronic HF.³⁴⁰ Waon also enhanced the favorable actions of RAAS inhibition and reduced SNS activity.³⁴¹

An additional strategy exploits upregulation of nNOS in the paraventricular nucleus and a reduction in sympathetic outflow. This observation raised the potential for nNOS upregulation by ACE2 therapy in HF.^{342,343}

ENDOTHELIN ANTAGONISTS

Acute ET nonselective antagonists decrease vascular resistance and increase cardiac index and cardiac output in HF. However, both short- and long-term clinical trials with ETA receptor antagonists have demonstrated no benefits but serious adverse events including fluid retention, increased need for hospitalization, hepatic dysfunction, and mortality in HF patients.³⁴⁴ These disappointing results may be explained by the observation that ET-A receptor antagonism in experimental HF further activates the RAAS in association with sustained Na⁺ retention.¹⁰⁸ Moreover, the increased cardiac, pulmonary, and renal ET-1 production in

HF, together with the marked vasoconstrictor and mitogenic properties of the molecule, suggests that ET-1 exacerbates Na^+ retention directly and indirectly by aggravating renal and cardiac functions, respectively.¹⁰⁸ In addition, nonspecific ET-B receptor blockade could have exacerbated cardiac remodeling. Given the antiproliferative effect of ET-B1–receptor activation and the detrimental vasoconstrictor effect of ET-B2 stimulation, combined ET-B1 stimulation and ET-B2 blockade has been proposed for control of the remodeling associated with postischemic HF.¹¹¹ Concomitant ET-A blockade would help to prevent the undesired effects of unopposed ET-1 activity and exploit the established therapeutic effects of ET-A antagonists.¹¹¹ In this regard, a report that dual ET-A/ET-B receptor inhibition improves echocardiographic parameters in a mouse model of HF with preserved ejection fraction (HfpEF) is encouraging.³⁴⁵ The favorable effects occurred by abrogating adverse cardiac remodeling and reducing stiffness. Additional studies on the role of dual ET-1 receptor antagonists in patients with HfpEF are eagerly awaited.

NATRIURETIC PEPTIDES

As noted previously, circulating NP levels are elevated in HF in proportion to disease severity, but renal actions of NPs are attenuated in severe HF. Nevertheless, elimination of NP action using NPR-A blockers or surgical removal of the atrium disrupted renal function and cardiac performance in experimental HF.³⁴⁶ Conversely, increasing circulating levels of NPs (BNP > ANP) by intravenous administration improved general clinical status³⁴⁷ and reduced pulmonary arterial and capillary wedge pressures, right atrial pressure, and SVR, with improved cardiac output, systemic BP, and diuresis. In parallel, plasma NE and aldosterone levels were suppressed.³⁴⁸ However, these beneficial effects have not translated into clinical efficacy, with several controlled studies showing minimal natriuretic effects of BNP (nesiritide) compared with placebo.⁹² Furthermore, serious adverse events of nesiritide treatment were seen including dose-related hypotension, worsening renal function, and possible increased risk of death.⁹² Finally, in a randomized controlled trial comparing nesiritide or low-dose dopamine as add-on therapy to intravenous furosemide in patients with HF and renal dysfunction, no improvement in decongestion or renal function was observed with either combination therapy over furosemide alone.³⁴⁹ Similarly, ularitide, the pharmacologic equivalent of the NP, urodilatin, as add-on therapy did not affect a clinical composite endpoint or reduce long-term cardiovascular mortality, despite significant reductions in NT-proBNP levels (TRUE-AHF trial). Therefore the therapeutic role of NP in HF remains unclear, and novel approaches are being explored. These include strategies to convert the prohormone into its active form, which might be efficacious; CNP analogs; and designer chimeric peptides such as CD-NP, consisting of CNP and part of DNP.³⁵⁰

NEUTRAL ENDOPEPTIDASE INHIBITORS AND VASOPEPTIDASE INHIBITORS

Correcting the imbalance between the RAAS and NP systems can also be achieved by inhibiting the enzymatic degradation of NPs by NEP. In both experimental models

and clinical trials, NEP inhibitors enhanced plasma ANP and BNP levels in association with vasodilation, natriuresis, diuresis, reduced cardiac preload, and afterload.¹³⁷ However, initial favorable hemodynamic, neurohormonal, and renal effects of NEP inhibitors in HF patients^{351,352} were not confirmed in later studies. These showed enhanced RAAS activation, increased ET-1 levels, and attenuated renal and hemodynamic improvement.¹³⁷ On the basis of these findings, combined RAAS and NEP (vasopeptidase) inhibitors were designed and, indeed, initially shown to have more favorable hemodynamic and renal effects, such as preserved GFR, than each treatment alone.¹³⁷

However, randomized clinical trials indicated that neither vasopeptidase inhibitors nor NEP inhibitors as add-on therapy to RAAS inhibition were more effective than RAAS inhibitors alone in the treatment of HF. Furthermore, the combination was associated with a significantly increased incidence of severe angioedema.¹³⁷ Possible reasons for the failure of NEP inhibitors include disproportionate increase in RAAS and ET activity over time, the development of tolerance with chronic treatment, and downregulation of NP receptors in response to degradation of NEP inhibitors. The increased incidence of angioedema with vasopeptidase inhibition may result from excessive accumulation of bradykinin or inhibition of aminopeptidase P.¹³⁷

The challenge of preventing angioedema led to development of combined ARB-NEP inhibition since ARBs do not disrupt bradykinin metabolism. Three trials in HF with reduced EF, IMPRESS, OVERTURE, and PARADIGM-HF, compared combined NEP/RAS inhibition with RAS inhibition alone and reported clinical outcomes. The pooled hazard ratio (HR) for all-cause death or HF hospitalization and all-cause mortality was significantly reduced in patients receiving combined NEP/RAS inhibition in all three trials. Combined NEP/RAS inhibition compared with ACE inhibition was associated with more hypotension but less renal dysfunction and hyperkalemia.³⁵³ These results pave the way for a paradigm shift in the management of severe HF.^{297,354}

VASOPRESSIN RECEPTOR ANTAGONISTS

AVP receptor antagonists (vaptans) are small, orally active, nonpeptide molecules that lack agonist effects and display high affinity for and specificity to their corresponding receptors.^{105,355} Highly selective and potent antagonists for the V_{1A} , V_2 , and V_{1B} receptor subtypes and mixed V_{1A}/V_2 receptor antagonists are now available.¹⁰⁵ In clinical trials and experimental models of compensated and decompensated HF with hyponatremia, V_2 receptor-specific vaptans produced hemodynamic improvement, with transient decrease in SVR and increased cardiac output. Water diuresis caused decreases in body weight and edema, hyponatremia was corrected, and RBF and renal function stabilized.^{105,355} Dyspnea lessened in some but not all patients. Both tolvaptan, a V_2 selective receptor antagonist, and conivaptan, a nonselective V_{1A}/V_2 receptor antagonist (both U.S. Food and Drug Administration [FDA] approved for use in HF) induced these favorable effects after treatment for up to 60 days. Similar results were reported for tolvaptan in stable class II or III HF.^{356,357} Positive effects were observed, regardless of whether LVEF was <40% or

>40%.^{105,355} In patients with LVEF <40%, a single dose of conivaptan reduced pulmonary capillary wedge and right atrial pressure in comparison with placebo. Cardiac index, pulmonary arterial pressure, systemic and pulmonary vascular resistance, systemic arterial pressure, and heart rate were unaffected. Urine output rose and osmolality fell significantly.^{358,359} However, functional capacity, exercise tolerance, and overall quality of life were unchanged.³⁵⁷

The effect of vaptans on survival is unclear. In one study, mortality rate 2 months after discharge was halved in patients treated with tolvaptan who had an increase in serum Na⁺ of 2 mmol/L or more, compared with those with no increase in serum Na⁺.³⁶⁰ In contrast, in the other trial, tolvaptan given for 60 days had no effect on survival after 9 months of follow-up.³⁵⁷ Lixivaptan, another V₂ selective antagonist, was associated with an excessive mortality rate, despite favorable clinical and laboratory effects similar to those of tolvaptan.³⁶¹ Hence the FDA did not approve lixivaptan for use in HF.³⁶²

Tolvaptan did not adversely affect cardiac remodeling or LVEF after 1 year of treatment in patients receiving optimal, background therapies for HF (RAAS and β -blockers).³⁶³ In contrast, in a murine model of HF, tolvaptan was associated with increased remodeling and mortality, which was prevented by concomitant furosemide therapy.³⁶⁴ Therefore vigilance will need to be exercised in the long-term use of tolvaptan in HF.

In view of the impressive diuretic effect of vaptans, there is considerable interest in a potential loop diuretic-sparing effect, as mentioned previously. To summarize results to date, tolvaptan compared with furosemide alone or in combination was not associated with improvement in dyspnea 24 hours after admission for acute HF, with or without preserved LVEF, despite greater diuresis and weight loss.³⁶⁵⁻³⁶⁸ Tolvaptan was associated with better preserved renal function and serum electrolytes.³⁶⁵⁻³⁶⁷ A Japanese multicenter clinical effectiveness of tolvaptan in patients with acute decompensated heart failure and renal failure (AQUAMARINE) study, in which tolvaptan was compared with standard furosemide therapy in acute HF patients with renal impairment, confirmed that early tolvaptan administration achieved more diuresis and relieved dyspnea symptoms.^{369,370}

The adverse effects of vaptans appear to be relatively few and, overall, minor. Thirst and dry mouth are not unexpected; hypokalemia occurs in <10% of recipients similar to loop diuretics. In the largest study to date, involving more than 4000 patients, there was a small but significant increase in strokes but a small but significant reduction in myocardial infarctions.⁹⁹

In summary, AVP receptor antagonists appear to have important short-term benefits in the treatment of advanced HF. However, the question of improvement in longer-term prognosis, especially the high mortality among HF patients already receiving optimal treatment with RAAS inhibitors ($\pm$ NEP inhibition) and β -blockers, remains open.³⁷¹

SODIUM-GLUCOSE TRANSPORTER-2 INHIBITORS

Sodium-glucose transporter-2 (SGLT-2) inhibitors decrease proximal tubular Na⁺ and Cl⁻ reabsorption, leading to a reset of TGF. This induces plasma volume contraction without

activation of the SNS, decreases glomerular hyperfiltration leading to better long-term renal preservation, and improves diuretic and natriuretic responses to other diuretics. Moreover, SGLT-2 inhibitors might improve the efficiency of myocardial energetics by offering β -hydroxybutyrate as an attractive fuel for oxidation and increase hematocrit improving oxygen transport. Finally, decreased vascular stiffness and improved endothelial function are observed with the use of SGLT-2 inhibitors in diabetes. These multiple nonglycemic effects make SGLT-2 inhibitors preferred glucose-lowering drugs in diabetic patients with HF, and its use has been approved in HF without diabetes.³⁷²⁻³⁷⁶ In a meta-analysis based on data from 9 RCTs involving 2824 patients, SGLT-2 inhibitor addition to conventional diuretic therapy for acute HF had reduced all-cause mortality, readmissions for HF, and the composite of cardiovascular death or readmissions for HF (odds ratio [OR] 0.75; 95% CI 0.56–0.99, $P = 0.049$; OR 0.54, 95% CI 0.44–0.66, $P < 0.001$; and hazard ratio 0.71, 95% CI 0.60–0.84, $P < 0.001$, respectively). Moreover, SGLT-2 inhibitor was associated with a higher volume of diuresis with a lower dose of loop diuretics.³⁷³ In the empagliflozin studies, clinical benefit was observed for both acute de novo and decompensated chronic HF and was observed regardless of ejection fraction or the absence of diabetes.^{297,374,375,377}

SPECIFIC TREATMENTS BASED ON THE PATHOPHYSIOLOGY OF SODIUM RETENTION IN CIRRHOSIS

The prognosis of type 1 HRS is dismal. The mortality rate, which rises with increased AKI stage, is as high as 80% in the first 2 weeks and only 10% of patients survive longer than 3 months. Therefore specific aggressive therapy in these patients is usually indicated in preparation for liver transplantation.^{184,237} Type 2 HRS has a better prognosis. Median survival is approximately 6 months.^{184,237} Aggressive management for HRS includes pharmacologic therapy, TIPS insertion, KRT, and liver transplantation. The management of precipitating factors is discussed elsewhere.

PHARMACOLOGIC TREATMENT

The goals of pharmacologic therapy are to reverse functional renal failure and serve as a bridge to liver transplantation. Earlier attempts to reverse renal vasoconstriction with low-dose dopamine, the PGE₁ analog misoprostol, RAAS, and ET inhibitors were unsuccessful and even detrimental.^{184,186,222} Therefore current treatment is based on reversal of splanchnic and systemic arterial vasodilation, AVP-induced water retention, and hyponatremia. Preventive therapy for ascites was studied in the β -blockers to prevent decompensation of cirrhosis in patients with clinically significant portal hypertension (PREDESCI) study.³⁷⁸ It showed that in patients with compensated cirrhosis and hepatic venous pressure gradient (HVPG) ≥ 10 mm Hg, long-term treatment with nonselective β -blockers improves decompensation-free survival, mainly by decreasing the incidence of ascites. Specifically, the benefit of β -blockers versus placebo in preventing ascites was slightly more apparent in the carvedilol stratum (HR 0.22, 95% CI

0.02–1.94) than in the propranolol stratum (0.50, 95% CI 0.22–1.18). Consequently, a competing-risk time-to-event meta-analysis using individual patient data obtained from principal investigators of recent RCTs confirmed that carvedilol is the treatment of choice. Low-dose carvedilol may also reduce HVPg in nonresponders to propranolol and, more importantly, carvedilol significantly enhances survival in compensated patients.³⁷⁹

SYSTEMIC VASOCONSTRICTORS

Three groups of vasoconstrictors have been studied—vasopressin V_1 receptor analogs, a somatostatin analog, and α -adrenergic agonists.^{184,237}

VASOPRESSIN V_1 RECEPTOR ANALOGS

These agents cause marked vasoconstriction through their action on V_1 receptors in the arterial smooth muscle wall. They are used for the management of acute variceal bleeding in cirrhosis and portal hypertension. Ornipressin infusion in combination with volume expansion led to a remarkable increase in RPF, GFR, and Na^+ excretion in almost 50% of treated patients. However, significant ischemic adverse effects in 30% of patients led to the abandonment of this drug.³⁸⁰

Terlipressin, without significant adverse ischemic reactions, is now the vasoconstrictor drug of choice. Terlipressin and albumin in type 1 HRS are associated with a significant improvement in GFR, increase in arterial pressure, near-normalization of neurohumoral levels, and reduction of serum creatinine level in 40% to 50% of cases.^{184,237,381} Response rates to terlipressin in type 2 HRS are better than those in type 1, with more than 80% survival at 3 months.^{182,240} Despite relapses in 50% of cases, reintroduction of therapy produces a further response.^{182,240}

The results of several randomized controlled trials, two meta-analyses, and several Cochrane reviews have shown that the combination of terlipressin and albumin is clearly superior to albumin or terlipressin alone in improving renal function and survival in both types 1 and 2 HRS and decreased the need for KRT, pretransplant, and posttransplant in type 1 HRS.³⁸²⁻³⁸⁶ No difference in efficacy between boluses and infusions was found, although the latter was associated with fewer adverse effects.³⁸⁷ The optimum duration of terlipressin therapy is not clear, but it is usually given until serum creatinine levels decrease to <1.5 mg/dL or for a maximum of 15 days. Whether extending therapy beyond 15 days will add any benefit is unknown. A key prognostic target after combined terlipressin-albumin therapy may be a rise in MAP of 10 mm Hg or more; these patients required less dialysis and were more likely to attain liver transplantation than those with smaller responses in MAP. This response was associated with better short-term, long-term, and transplant-free survival.³⁸⁸ Moreover, pretransplantation normalization of renal function by terlipressin resulted in similar posttransplantation outcomes to those of patients with normal renal function.³⁸⁹

In patients with HRS undergoing living-related donor liver transplantation, terlipressin intraoperatively and postoperatively is associated with significant increases in MAP, SVR, and renal function. Significantly decreased heart rate, cardiac output, hepatic and renal arterial resistive indices, portal venous blood flow, and use of vasoconstrictor

drugs were seen during reperfusion.^{390,391} To summarize, terlipressin and albumin infusion may be appropriate only for patients awaiting or already undergoing liver transplantation and a favorable hemodynamic response is associated with a better overall prognosis both before and after liver transplantation.

CONFIRM (A Multi-Center, Randomized, Placebo Controlled, Double-Blind Study to Confirm Efficacy and Safety of Terlipressin in Subjects With Hepatorenal Syndrome Type 1) was the third and largest RCT evaluating terlipressin based in North America. It had stringent criteria for eligibility and “verified” HRS reversal. The study met its primary endpoint with 63 of 199 (32%) patients in the terlipressin arm as compared with 17 of 101 (17%) in the placebo arm who achieved the verified HRS reversal ($P = 0.006$).³⁹² However, the study raised concerns about respiratory adverse events and a lack of evidence for mortality benefit. Until now, no randomized trial investigating terlipressin for HRS-AKI was powered to assess its impact on mortality.³⁹³ In September 2022, the FDA granted approval to terlipressin for the treatment of hospitalized adults with HRS and rapid decline in kidney function.

SOMATOSTATIN ANALOGS AND α -ADRENERGIC AGONISTS

Octreotide is an α -adrenergic agonist that inhibits the release of glucagon and other vasodilator peptides. Octreotide with albumin infusion or midodrine had no effect on renal function in HRS. However, both agents in combination with albumin infusion led to a significant improvement in renal function and survival in types 1 and 2 HRS in comparison with controls.^{394,395,396} However, no effect of this combination on outcomes of subsequent liver transplantation was observed.³⁹⁷ Attempts to prevent relapse of type 2 HRS with midodrine after terlipressin-induced improvement were also unsuccessful.³⁹⁸ Moreover, terlipressin plus albumin was significantly more effective than octreotide, midodrine, and albumin in terms of improved renal function, complete reversal of HRS, and 90-day survival.³⁹⁹ Finally, NE was compared with terlipressin in the treatment of type 1 HRS in three randomized controlled trials. Responses to both agents were similar in terms of MAP and renal function. Cumulative survival and adverse event rates were not significantly different between the two drugs. NE is less expensive than terlipressin, but, unlike terlipressin, NE administration requires cardiac monitoring. Therefore total costs might be similar for the two therapies.^{184,237}

VASOPRESSIN V_2 RECEPTOR ANTAGONISTS

As noted, hyponatremia, caused by persistent nonosmotic AVP-induced water retention, is often seen in HRS and is a marker of poor prognosis.⁴⁰⁰ Therefore attaining a water diuresis and reversing hyponatremia using V_2 receptor antagonists are potentially important therapeutic goals. Several controlled studies using lixivaptan, tolvaptan, and satavaptan have shown modest improvements in serum Na^+ , but without effect on mortality or adverse events.^{400,401,402} Satavaptan in combination with diuretics was associated with a higher mortality rate than either agent alone, leading to early termination of one study.⁴⁰³ The specific role of satavaptan in the increased mortality was uncertain, given

that most deaths were due to complications of cirrhosis. Overall, the effects of vaptans in HRS appear to be modest; this may be explained by avid proximal tubular solute reabsorption leading to reduced distal delivery or by V_2 receptor-independent pathways of water retention.¹⁰³

TRANSJUGULAR INTRAHEPATIC PORTOSYSTEMIC SHUNT

The efficacy of TIPS in the reduction of portal venous pressure in patients with cirrhosis and refractory ascites with type 1 or 2 HRS has been demonstrated in several small cohort and controlled studies.^{182,404} Significant improvement in renal hemodynamics, GFR, and vasoconstrictive neurohumoral factors were observed in most patients.²²² Liver transplant-free survival also was markedly improved⁴⁰⁴ and might be better than that reported for terlipressin and albumin infusion.⁴⁰⁵ Some patients who were dialysis dependent were able to discontinue this treatment after TIPS insertion. Moreover, liver transplantation was performed in two patients when the medical condition that precluded transplantation had resolved after TIPS insertion.²⁷⁹ However, TIPS seems to increase wait time for liver transplantation and days in hospital after transplantation, although without effect on 30-day mortality.⁴⁰⁶

TIPS appears to exert its favorable effects by reducing sinusoidal hypertension with suppression of the putative hepatorenal reflex (see earlier), improvement of EABV by shunting portal venous blood into the systemic circulation, or amelioration of cardiac dysfunction.²⁵³

TIPS is currently recommended in the setting of refractory ascites, particularly in type-2 HRS, but is contraindicated in severe liver failure, severe intrinsic renal failure (serum creatinine >3 mg/dL), HF, severe portopulmonary hypertension, recurrent or persistent severe hepatic encephalopathy despite adequate treatment, and uncontrolled sepsis.^{287,289,407}

KIDNEY AND LIVER REPLACEMENT THERAPY

The benefits of conventional hemodialysis and continuous KRT, in terms of prolonging survival, are dubious,⁴⁰⁸ and morbidity resulting from these therapies is high.⁴⁰⁹ In oliguric patients awaiting liver transplantation who do not respond to vasoconstrictors or TIPS and who develop diuretic-resistant volume overload, hyperkalemia, or intractable metabolic acidosis, KRT may be considered. In view of the dismal prognosis of HRS, especially type 1, KRT should probably be used solely as a bridge to transplantation.

In contrast with conventional KRT, albumin dialysis offers the potential advantage of removing albumin-bound, water-soluble vasoactive agents, toxins, and proinflammatory cytokines. Molecules relevant to the pathogenesis of advanced cirrhosis include bile acids, TNF- α , interleukin-6, and NO.¹⁸⁶ Three albumin dialysis systems, molecular adsorbent recirculating system (MARS), fractionated plasma separation, adsorption and hemodialysis (Prometheus system), and single-pass albumin dialysis, have been examined in randomized trials for supportive treatment of liver failure. These therapies enable partial recovery of hepatic function and provide simultaneous KRT. They lead to a decrease in renal vascular resistance and improvement in the splenic resistance index, a parameter related to portal

resistance. The hemodynamic effects are thought to be mediated by clearance of vasoactive substances.⁴¹⁰ Bilirubin levels, encephalopathy grade, serum creatinine, and serum Na^+ all improved.⁴¹¹ In one meta-analysis, albumin dialysis achieved a significant net decrease in serum total bilirubin level relative to standard medical therapy but not in serum ammonia or bile acids. Albumin dialysis achieved an improvement in hepatic encephalopathy relative to standard medical therapy but had no effect on survival. Bearing in mind the limited reporting of adverse events, no major safety concerns were detected. Extracorporeal nonbiologic liver support system should at present probably be considered only as a bridge to transplantation.⁴¹²

LIVER TRANSPLANTATION

Liver transplantation is the treatment of choice for HRS because it offers a cure for liver disease and renal dysfunction. Large case series indicate that survival is significantly lower in transplant recipients with HRS (types 1 and 2) both immediately postoperatively and long term than in those without the syndrome (3-year survival rates of 60% vs. 70% to 80%) but may be improved by the bridging therapies described previously.^{413,414}

With respect to renal function after transplantation in patients with HRS, GFR decreases in the first month owing to the stress of surgery, infections, immunosuppressive therapy, and other factors. Despite the prompt correction of hemodynamic and neurohumoral parameters, GFR recovers incompletely (30–40 mL/min at 1–2 months) and often remains impaired over the long term. Although dialysis requirement in the first month is greater in patients with HRS than without HRS (35% vs. 5%), duration of dialysis pretransplantation does not influence renal recovery post transplantation. Predictors of renal recovery included younger recipient and donor, nonalcoholic liver disease, and low posttransplantation serum bilirubin.⁴¹⁵ Overall, the rate of posttransplantation reversal of HRS has been estimated to be no greater than 58%.

Unsurprisingly, the 5-year risk of ESRD post transplantation in patients with HRS is significantly higher than in those without HRS. Moreover, combined simultaneous liver and kidney transplantation (SLKT) apparently offers no greater benefit over liver transplantation alone with respect to early posttransplantation kidney function. However, better 5-year survival rates are more likely after SLKT than after liver transplantation alone in patients with pretransplantation serum creatinine levels >2.2 mg/dL, irrespective of dialysis requirement.⁴¹⁴ More studies are needed to enable a rational decision about whom should receive SLKT, rather than liver transplants alone. At present, SLKT should probably be reserved for patients who are dialysis dependent for 8 weeks or more pretransplantation.^{186,414,416}

The introduction of Model of End-stage Liver Disease (MELD) scores for the allocation of livers has increased the number of transplantations in patients with impaired renal function, but more SLKTs are also being performed.⁴¹⁷ On the other hand, a favorable response to vasoactive therapy reduces the MELD score and may lead to a paradoxical delay in liver transplantation.⁴¹⁸ Therefore only pretreatment MELD scores should be used to predict potential outcomes of liver transplantation in HRS.¹⁸² A further paradox is that

patients with type 2 HRS have lower MELD scores than those with type 1 HRS, resulting in the former being ascribed a lower priority for transplantation and a longer time on the waiting list. This delay is associated with higher mortality, especially in the presence of hyponatremia and persistent ascites.⁴¹⁷ Therefore the criteria for donor allocation need to be modified to incorporate these factors into the final score for prioritization.¹⁸²

For a general summary of all aspects of liver transplantation, the reader is referred to excellent reviews.^{419,420}

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